THE UNIVERSITY OF HONG KONG
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How Do Parental Socioeconomic Resources Shape Children's Occupational Outcomes?
A Quantile-Based Analysis Using CFPS (2010–2022)

A capstone project report submitted in partial fulfillment for the degree of Master of Social Sciences in the field of Population and Policy Analysis

By

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Declaration

We declare that this capstone project report represents our group's own work, except where due acknowledgement is made, and that it has not been previously included in a thesis, dissertation or report submitted to this University or to any other institutions for a degree, diploma or other qualification.

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Acknowledgements

Abstract

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Chapter 1

Introduction

1.Intro and Research Background

1.1 Research Background

The intergenerational transmission of socioeconomic status (SES) is a central theme in social stratification and mobility research. A fundamental question in this field is whether an individual's career success is a result of their own merit (educational attainment) or a reflection of their "ascribed" characteristics, such as family background. In modern societies, while education is often viewed as the primary engine for social mobility, empirical evidence suggests that the "long shadow" of the family still plays a decisive role in shaping the career trajectories of the next generation.

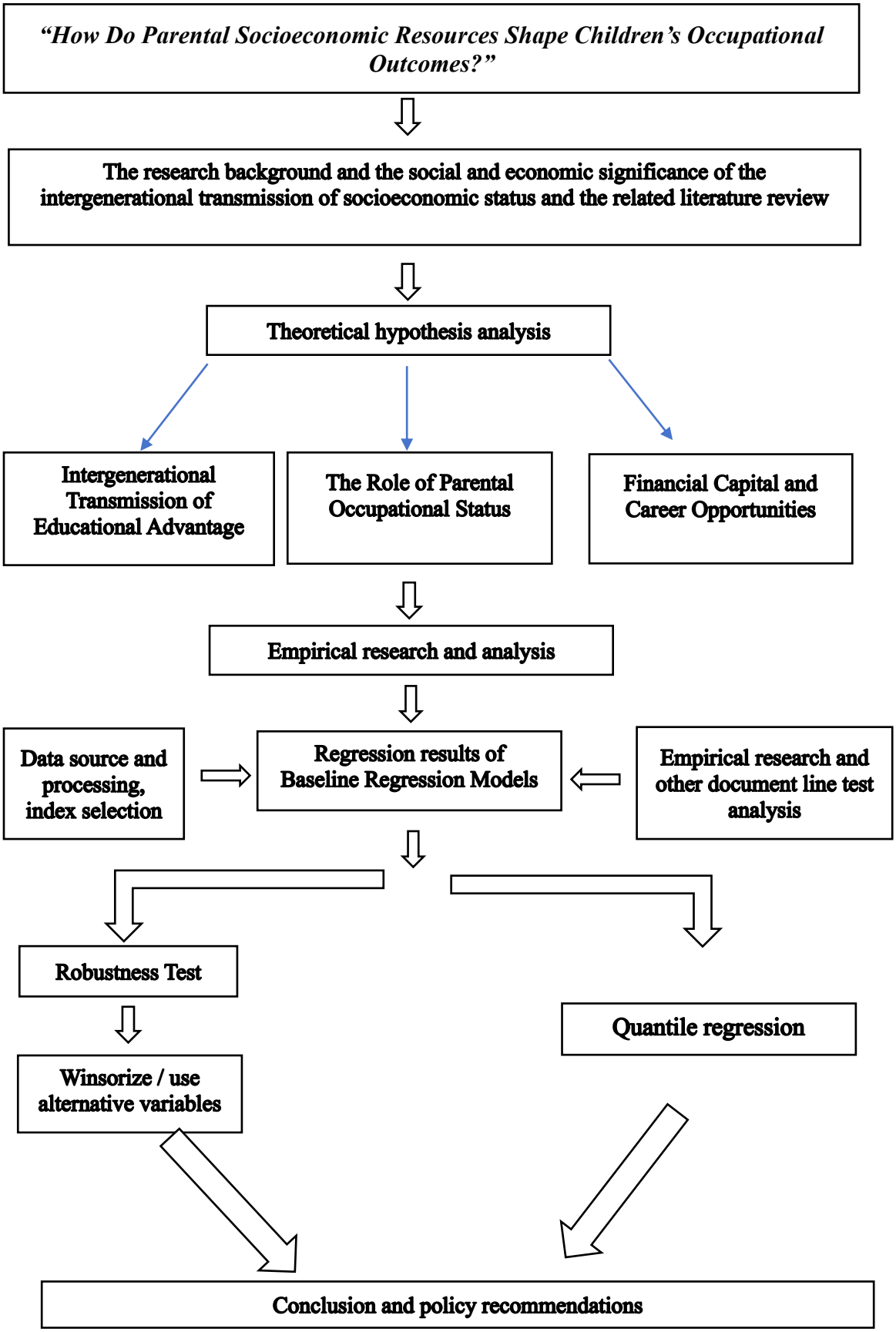
In the context of contemporary China, this issue is particularly salient. Over the past few decades, China has experienced rapid economic transformation and a massive expansion of its higher education system. While these changes have created more opportunities for upward mobility, they have also reshaped the mechanisms through which parents pass on their advantages to their children. The persistent divide between rural and urban areas, coupled with the rising importance of social and cultural capital, has made the link between parental SES and children’s occupational outcomes more complex and multifaceted.

1.2 The Research Problem

Despite the growing body of literature on status attainment, several gaps remain. First, how do different dimensions of parental SES—such as education, occupational prestige (ISEI), and family income—interact to influence a child's career choice? Second, in a transitioning economy like China, does the family's economic capital (income) outweigh their cultural capital (education) in securing high-status jobs for their children?

This study utilizes data from the China Family Panel Studies (CFPS) to examine the impact of parental SES on children’ s occupational choices. By analyzing variables such as maternal and paternal education, family income, and parental ISEI, we aim to quantify the extent of intergenerational persistence. Specifically, we investigate how parental resources facilitate children's entry into high-prestige occupations and whether this influence varies across gender and residential (rural/urban) lines.

1.3 Research Flowchart



Chapter 2

Research Background

Chapter 3

Literature Review

2. Literature Review

2.1. Intergenerational Advantage and Distributional Heterogeneity

The intergenerational transmission of economic advantage has been a central concern in social science research, with scholars examining how parental wealth shapes children's educational attainment, labor market outcomes, and health across the life course. Recent advances in quantile regression methods have enabled researchers to uncover heterogeneous effects that vary across the outcome distribution, revealing that parental resources may amplify advantages particularly at the upper tail of children's achievement distributions.

2.2. Human Capital Theory and Intergenerational Mobility

Human capital theory and intergenerational mobility provide the foundational framework for understanding parental wealth effects. Araki (2025) demonstrates that education enhances productivity and leads to better labor market performance, traditionally captured by the Mincer earnings function relating years of schooling to wages (Social Science & Medicine, 381, p. 2). This framework is particularly relevant when considering how parental wealth enables educational investments. The quantile regression analysis in Araki's study reveals substantial heterogeneity in returns to college completion across the wage distribution, with direct effects at age 40 ranging from 0.26 at the mean to 0.64 at the 90th percentile (p. 7), suggesting that children from wealthier families who reach higher education levels experience disproportionately larger economic returns.

2.3. The Education – Health Gradient and Cultural Health Capital

The education-health gradient represents a critical pathway through which parental wealth influences children's long-term well-being. The literature presents mixed evidence on causal effects of education on health outcomes. As documented in the comprehensive review, studies using compulsory schooling reforms as natural experiments show divergent findings: while Gathmann, Jürges, and Reinhold (2015) found positive effects on mortality across twentieth-century Europe, other studies including Arendt (2005) for Denmark, Albouy and Lequien (2009) for France, and Meghir et al. (2018) for Sweden found no causal effect (Education and Adult Health, 2020, p. 3). This inconsistency suggests that the education-health relationship may be contingent on institutional contexts and family resources. Hamad et al. (2018) conducted a systematic review and meta-analysis specifically examining studies of compulsory schooling laws, finding that methodological differences and contextual factors explain much of the disagreement in the literature (p. 168-178). The concept of "cultural health capital" introduced by Dubbin, Chang, and Shim (2013) further illuminates how parental socioeconomic status shapes health-related resources and patient-provider interactions (Social Science & Medicine, 93, p. 113-120), indicating that wealth effects extend beyond financial resources to include social and cultural competencies.

2.4. Quantile Regression and RIF Regressions

Quantile regression methodology has emerged as essential for capturing the distributional heterogeneity in parental wealth effects. Firpo, Fortin, and Lemieux (2018) developed recentered influence function (RIF) regressions for decomposing wage distributions, enabling researchers to

estimate effects at different quantiles rather than relying solely on mean effects (Econometrics, 6(2), p. 1-40). Araki (2025) applies this methodology to demonstrate that both direct and indirect effects of education on wages vary substantially across quantiles, with the ratio of indirect (health-mediated) to total effects being larger at higher quantiles (Q90) than at lower quantiles (Q10). This finding implies that children from wealthy families who achieve higher education not only earn more directly but also benefit more from health-related productivity advantages, creating a compounding effect of initial family wealth.

2.5. Labor Market Mechanisms

SkilLls, tasks, and technological change shape how parental investments translate into labor market returns. The literature emphasizes that task measures collected at the individual worker level offer substantial additional explanatory power for wages relative to occupation-level data. Studies using the German IAB/BIBB survey, including work by Spitz-Oener (2006), Dustmann et al. (2009), and Gathmann and Schönberg (2010), demonstrate that individual task measures provide crucial insights into skill formation and wage determination (Skills, Tasks and Technologies, p. 36). Autor and Handel (2009) similarly show that individual task measures from the PDII survey instrument offer substantial additional explanatory power for wages relative to occupation-level data. This research suggests that parental wealth may influence not just educational credentials but also the specific skills and tasks children are able to develop and deploy in the labor market, with implications for understanding inequality reproduction across generations.

2.5. New Challenges to Mobility

Emerging health disparities and generational decline present concerning trends for intergenerational mobility. Zheng, Dirlam, Choi, and George (2023) documented the health decline of Americans from baby boomers to millennials, suggesting deteriorating intergenerational outcomes despite rising educational attainment (Social Science & Medicine, 337, article 116282). Zheng, Lu, and Yao (2024) identified emerging health disparities specifically among college graduates, highlighting education-occupation mismatch as a mechanism that may undermine the health benefits traditionally associated with higher education (Social Science Research, 120, article 103015). These findings raise important questions about whether parental wealth can continue to buffer children against broader societal trends of declining health and economic security. The work by Easterlin and colleagues (1990, 1993) on baby boomer economic well-being provides historical context, showing that even advantaged cohorts may face challenges in matching their parents' economic status when entering labor markets during periods of structural change (Journal of Population Economics, 3(4), p. 277-290; Population and Development Review, 19(3), p. 497-522).

2.6. Causal Inference in Wealth-Education-Health Links

Methodological considerations and causal identification remain central challenges in this literature. The heterogeneity in findings regarding education's causal effect on health underscores the difficulty of isolating parental wealth effects from confounding factors. Studies using twin data, such as Fujiwara and Kawachi (2009), Amin et al. (2015), and Sudharsanan et al. (2016), generally find small or no effects of education on health, suggesting that family background factors shared by twins may be more important than educational differences per se (Education and Adult Health, 2020, p. 3). This implies that observed correlations between parental wealth, children's education, and children's outcomes may partly reflect genetic or family environmental factors rather than purely causal effects of wealth-enabled investments. The application of quantile regression methods helps address this concern by revealing whether effects are concentrated at particular points in the outcome distribution where selection effects might be less severe.

2.7. Bibliometric Research

Furthermore, this study also employed a bibliometric approach to examine the research landscape of social mobility. The data were collected from the Web of Science Core Collection (WoSCC). The search was conducted using the query TS = ("social mobility" OR "intergenerational mobility"). In WoS, TS refers to Topic Search, which covers the title, abstract, author keywords, and Keywords Plus.

Figure 1 shows the top 10 most frequent keywords in the dataset. Social mobility is the most frequent keyword, followed by education and intergenerational mobility. This indicates that the field is primarily concerned with mobility processes, educational factors, and intergenerational transmission. Other frequently occurring keywords, such as family background, social stratification, educational attainment, and occupational attainment, suggest that the literature is closely related to inequality, class reproduction, and status attainment.

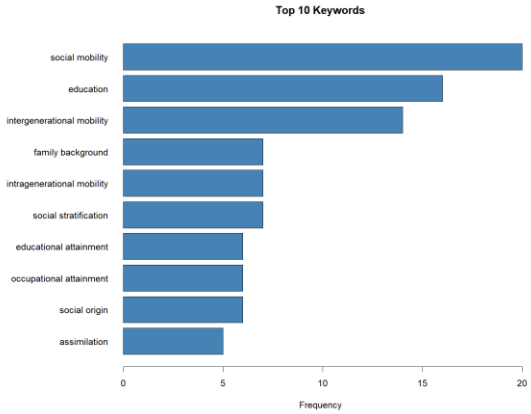


Figure 2 presents the keyword co-occurrence network. The largest nodes, including inequality, social mobility, attainment, and united-states, occupy central positions in the network, indicating their importance in the field. The network also shows strong connections among family background, education, occupational attainment, and intergenerational mobility, suggesting that these topics form the core conceptual structure of the literature. In addition, keywords such as childhood, children, earnings, and work indicate a related focus on life-course outcomes and labor market consequences.

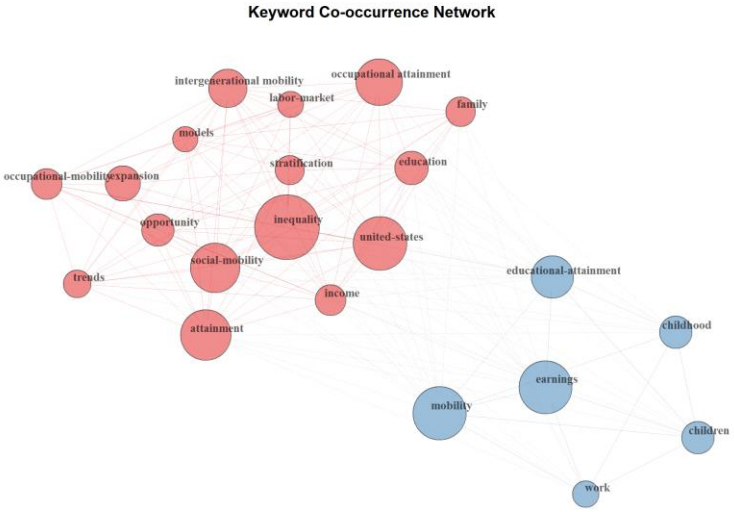


Figure 3 illustrates the country collaboration network. The United States is the most central and influential country in the network, with strong collaborative ties to the United Kingdom, Germany, and Sweden. This suggests that the field is largely dominated by the United States and several European countries. Although other countries, such as Canada, the Netherlands, Finland, China, and Switzerland, also participate in international collaboration, their positions are relatively less central. Overall, the network reveals an uneven but active pattern of global academic cooperation.

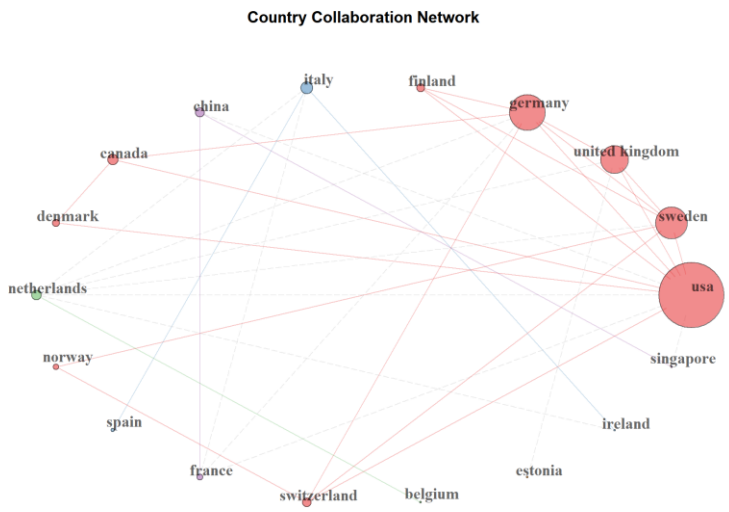
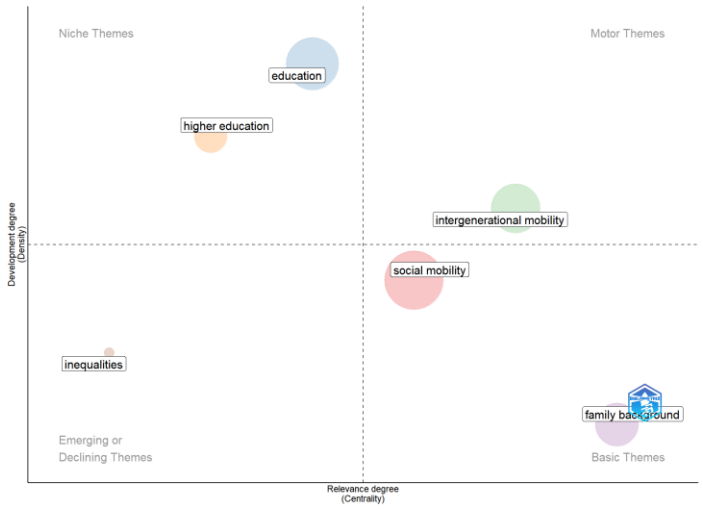


Figure 4 presents the thematic structure of the field. Intergenerational mobility appears in the motor-theme area, indicating that it is both important and relatively well developed. Social mobility and family background are located in the basic-theme area, showing that they are central to the field but remain broad and foundational topics. Education and higher education are positioned in the niche-theme area, suggesting that they are internally developed but less central to the overall field. Inequalities appears in the emerging or declining theme area, indicating that it is either still developing or less consolidated in the current sample.



Overall, the bibliometric results show that the literature on social mobility is mainly structured around education, family background, inequality, and intergenerational mobility. The field is supported by active international collaboration, although research output is concentrated in a limited

number of countries, especially the United States and Western Europe. These findings suggest that social mobility research continues to focus on how structural inequality and family-related factors shape educational, occupational, and life-course outcomes.

Chapter 4

Theory and Research Hypothesis

This study is grounded in theories of intergenerational mobility, human capital accumulation, and social stratification, which emphasize the persistence of socioeconomic advantage across generations. Parents influence their children’s labor-market outcomes through multiple, interrelated channels, including educational investment, occupational role modeling, and the provision of financial and social resources. These mechanisms jointly shape children’s skill formation, educational trajectories, and eventual occupational attainment.

3.1 Intergenerational Transmission of Educational Advantage

Education constitutes a central mechanism through which socioeconomic status is transmitted across generations. According to human capital theory, parental education enhances children’s cognitive and non-cognitive skill development by shaping early learning environments, educational aspirations, and academic support. Highly educated parents are more likely to possess the cultural capital necessary to navigate educational institutions, transmit effective learning strategies, and set higher expectations for academic achievement. These advantages translate into greater educational attainment for children, which in turn increases access to high-status occupations.

Moreover, education not only improves formal qualifications but also fosters problem-solving abilities, communication skills, and adaptability—traits that are increasingly rewarded in modern labor markets. Consequently, children from more educated families are better positioned to sort into occupations with higher prestige and socioeconomic status.

Hypothesis 1 (Educational Transmission):
Parental educational attainment has a significant and positive effect on children’s occupational outcomes.

3.2 Parental Occupational Status and Social Capital

Beyond education, parental occupational status reflects accumulated labor-market advantages such as professional reputation, workplace experience, and access to valuable social networks. Parents employed in high-status occupations are more likely to transmit occupation-specific knowledge, provide career-related information, and facilitate entry into prestigious professions through referrals and informal connections. These forms of social capital are particularly important in labor markets where hiring practices rely on networks and reputation rather than purely formal credentials.

In addition, children may internalize occupational norms and aspirations through parental role modeling, shaping their career expectations and job search behavior. Parental occupational status therefore influences not only opportunities but also preferences, reinforcing intergenerational persistence in occupational stratification.

Hypothesis 2 (Occupational Status Transmission):

Parental occupational status, measured by ISEI, is positively associated with the prestige of children's occupational outcomes.

3.3 Financial Capital and Career Opportunities

Family income represents a critical resource that relaxes budget constraints over the life course. Higher-income households can invest more in children's education through better schooling, tutoring, extracurricular activities, and geographic mobility. Financial resources also reduce children's exposure to economic uncertainty, allowing them to pursue longer educational pathways, accept unpaid or low-paid internships, and delay labor-market entry in favor of skill accumulation.

From a life-cycle perspective, credit constraints disproportionately affect children from low-income families, forcing earlier labor-market participation and limiting access to high-return educational investments. As a result, family income plays an independent role in shaping occupational outcomes, even after accounting for parental education and occupational status.

Hypothesis 3 (Financial Capital Effect):
Higher family income is associated with more favorable occupational outcomes for children.

3.4 Theoretical Framework

This study examines how parental socioeconomic status (SES) shapes children's occupational attainment. The literature on intergenerational mobility suggests that family background affects children's later-life outcomes through multiple channels. On the one hand, parental SES influences children's educational attainment by shaping access to educational resources, learning environments, aspirations, and institutional opportunities. On the other hand, parental SES may also exert a direct effect on children's occupational attainment through social networks, cultural capital, family expectations, information advantages, and differen-tial access to desirable occupations.

To formalize these mechanisms, this section develops a simple status attainment frame-work in which parental SES affects children's occupational outcomes both indirectly through education and directly through non-educational channels. The framework also allows the strength of this effect to vary across the occupational distribution, thereby providing a the-oretical motivation for the quantile regression analysis in the empirical section.

3.4.1 A Two-Stage Status Attainment Model

Let P_i denote parental socioeconomic status for individual i, E_i denote the individual's educational attainment, and O_i denote the individual's occupational attainment. Let X_i be a vector of observed individual and family characteristics, such as gender, cohort, hukou status, or other background controls. We assume that occupational attainment is generated through a two-stage process.

In the first stage, children's educational attainment is determined by parental SES and other background characteristics:

$$E_i = \alpha_0 + \alpha_1 P_i + X_i' \alpha_2 + u_i \tag{1}$$

where u_i is an error term capturing unobserved determinants of educational attainment.

In the second stage, children's occupational attainment depends on both their own educational attainment and parental SES:

$$O_i = \beta_0 + \beta_1 E_i + \beta_2 P_i + X_i' \beta_3 + \varepsilon_i \quad (2)$$

where ε_i is an error term capturing unobserved determinants of occupational attainment.

Equation (1) captures the idea that children from more advantaged families tend to achieve higher levels of education. Equation (2) reflects the standard status attainment logic that education raises occupational status, while parental SES may retain an independent effect even after education is controlled for.

3.4.2 Interpretation of the Structural Parameters

The structural coefficients in equations (1) and (2) have straightforward substantive interpretations.

First, α_1 measures the effect of parental SES on children's educational attainment. A positive value of α_1 implies that parental advantage is transmitted into educational advantage.

Second, β_1 measures the occupational return to education. A positive value of β_1 means that higher educational attainment is associated with higher occupational status.

Third, β_2 measures the direct effect of parental SES on occupational attainment after conditioning on education. This direct effect may reflect non-meritocratic mechanisms of status transmission, including family networks, social closure, occupational inheritance, and other forms of advantage not fully mediated by schooling.

Together, these parameters describe how intergenerational stratification may persist even in contexts where education appears to be the primary mechanism of mobility.

3.4.3 Reduced-Form Occupational Attainment Equation

By substituting equation (1) into equation (2), we can derive a reduced-form equation that links parental SES directly to children's occupational attainment.

Substituting for E_i yields:

$$\begin{aligned} O_i &= \beta_0 + \beta_1(\alpha_0 + \alpha_1 P_i + X_i{}' \ \alpha_2 + u_i) + \beta_2 P_i + X_i'^{\beta_3} + \varepsilon_i \quad (3) \\ &= (\beta_0 + \beta_1\alpha_0) + (\beta_1\alpha_1 + \beta_2)P_i + X_i'^{(\beta_1\alpha_2+\beta_3)} + \beta_1 u_i + \varepsilon_i \end{aligned}$$

Define

$$\pi_0 = \beta_0 + \beta_1\alpha_0 \quad (4)$$

$$\pi_1 = \beta_1\alpha_1 + \beta_2 \quad (5)$$

$$\pi_2 = \beta_1\alpha_2 + \beta_3 \quad (6)$$

and

$$\eta_i = \beta_1 u_i = \varepsilon_i \quad (7)$$

Then the reduced-form occupational attainment equation becomes

$$O_i = \pi_0 + \pi_1 P_i + X_i' \pi_2 + \eta_i \quad (8)$$

Equation (8) provides the empirical basis for estimating the overall association between

parental SES and children's occupational attainment. Importantly, the reduced-form coefficient π_1 incorporates both the indirect effect of parental SES operating through education and the direct effect of parental SES operating through other channels.

3.4.4 Proposition 1: Total Effect of Parental SES

We now characterize the total effect of parental SES on children's occupational attainment.

Proposition 1. *Under equations (1) and (2), the total marginal effect of parental SES on children's occupational attainment is*

$$\frac{\partial O_i}{\partial P_i} = \beta_1\alpha_1 + \beta_2 \tag{9}$$

Proof. From equation (2),

$$O_i = \beta_0 + \beta_1 E_i + \beta_2 P_i + X_i'\beta_3 + \varepsilon_i \tag{10}$$

Because E_i depends on P_i through equation (1), the total derivative of O_i with respect to P_i is

$$\frac{\partial O_i}{\partial P_i} = \beta_1 \frac{\partial E_i}{\partial P_i} + \beta_2 \tag{11}$$

From equation (1), we have

$$\frac{\partial E_i}{\partial P_i} = \alpha_1 \tag{12}$$

Substituting this result into equation (11) yields

$$\frac{\partial O_i}{\partial P_i} = \beta_1\alpha_1 + \beta_2 \tag{13}$$

This proves the proposition.

Proposition 1 shows that the effect of parental SES on occupational attainment does not operate through a single channel. Instead, the total effect is the sum of an indirect component transmitted through educational attainment and a direct component that remains after controlling for education.

3.4.5 Decomposition into Indirect and Direct Effects

The total effect derived above can be decomposed more explicitly.

Proposition 2. *The total effect of parental SES on children's occupational attainment can be decomposed as*

$$\textit{Total Effect} = \textit{Indirect Effect} + \textit{Direct Effect} \tag{14}$$

where

$$\textit{Indirect Effect} = \beta_1\alpha_1 \tag{15}$$

and

$$\textit{Direct Effect} = \beta_2 \tag{16}$$

Proof. From Proposition 1,

$$\frac{\partial O_i}{\partial P_i} = \beta_1\alpha_1 + \beta_2 \tag{17}$$

By construction, α_1 measures how parental SES affects education, and β_1 measures how education affects occupational attainment. Therefore, the product $\beta_1\alpha_1$ captures the effect of parental SES on occupational attainment that is mediated by education. The residual termβ_2 captures the direct effect of parental SES not transmitted through education. Hence, the total effect can be written as the sum of the indirect effect and the direct effect.

The decomposition in Proposition 2 is substantively important. If the indirect effect is dominant, then education is the principal mechanism through which family background shapes occupational outcomes. If the direct effect remains large, then family advantage persists even conditional on education, suggesting that occupational stratification is reproduced through channels beyond schooling.

3.4.6 Amplification through the Return to Education

The framework also implies that the magnitude of intergenerational transmission depends on the occupational return to education.

Proposition 3. *Suppose α1 > 0. Then an increase in the occupational return to education, β_1, increases both the indirect effect of parental SES and the total effect of parental SES on children's occupational attainment.*

Proof. From equation (15), the indirect effect is

$$I = \beta_1\alpha_1 \tag{18}$$

Differentiating with respect to β_1 gives

$$\frac{\partial I}{\partial \beta_1} = \alpha_1 \tag{19}$$

If $\alpha_1 > 0$, then

$$\frac{\partial I}{\partial \beta_1} > 0 \tag{20}$$

Thus, a higher occupational return to education magnifies the indirect effect of parental SES. Similarly, from equation (9), the total effect is

$$T = \beta_1\alpha_1 + \beta_2 \tag{21}$$

Differentiating with respect to β1 gives

$$\frac{\partial T}{\partial \beta_1} = \alpha_1 \tag{22}$$

Again, if $\alpha_1 > 0$, then $\partial T/\partial\beta_1 > 0$. Therefore, a higher return to education also increases the total effect of parental SES.

Proposition 3 highlights an important implication: educational expansion does not necessarily weaken intergenerational inequality. If access to education remains socially stratified and the labor market strongly rewards educational credentials, then a rise in the return to education may increase, rather than decrease, the role of family background in shaping occupational attainment.

3.4.7 Conditions for Meritocratic Attainment

The model also clarifies the conditions under which occupational attainment would be more consistent with a meritocratic process.

Proposition 4. *Occupational attainment is conditionally meritocratic with respect to parental SES if and only if* $\beta_2 = 0$*. In this case, the effect of parental SES on occupational attainment operates entirely through education.*

Proof. From equation (9),

$$\frac{\partial O_i}{\partial P_i} = \beta_1\alpha_1 + \beta_2 \tag{23}$$

The direct contribution of parental SES to occupational attainment is captured by β_2. If $\beta_2 = 0$, then

$$\frac{\partial O_i}{\partial P_i} = \beta_1\alpha_1 \tag{24}$$

which means that parental SES affects occupational attainment only through education. Conversely, if $\beta_2 \neq 0$, then parental SES retains an effect net of education, implying that occupational attainment is influenced by family background through non-educational channels. Therefore, conditional meritocracy requires $\beta_2 = 0$.

This proposition does not imply that a society is fully meritocratic whenever $\beta_2 = 0$, since educational attainment itself may already embody substantial family inequality. Rather, it means that, conditional on education, parental SES no longer provides an additional advantage in occupational attainment.

3.4.8 Distributional Heterogeneity in Occupational Attainment

The preceding analysis focuses on average effects. However, the influence of parental SES may vary across the distribution of occupational attainment. This possibility is especially relevant in studies of social stratification because access to top occupations may depend more heavily on family resources than access to middle- or lower-status occupations.

To allow for such heterogeneity, consider the following conditional quantile specification:

$$Q_\tau(O_i \mid P_i, X_i) = \delta_{0\tau} + \delta_{1\tau}P_i + X_i'\delta_{2\tau} \tag{25}$$

where $Q_\tau(O_i \mid P_i, X_i)$ denotes the τ-th conditional quantile of occupational attainment.

In this setting,$\delta_1\tau$ measures the marginal effect of parental SES at quantile τ of the conditional occupational distribution.

Proposition 5. *If $\delta 1\tau$ is increasing in τ , then parental SES has a stronger effect among children located at higher points of the occupational attainment distribution.*

Proof. Take any two quantiles τ_a and τ_b such that $\tau_b > \tau_a$. If

$$\delta_{1\tau_b} > \delta_{1\tau_a}, \tag{26}$$

then the effect of parental SES on the conditional quantile of occupational attainment is larger at τ_b than at τ_a. Therefore, parental SES exerts a stronger influence among individuals at higher positions of the occupational distribution.

This proposition provides the theoretical rationale for estimating quantile regressions in addition to mean regressions. If family advantage is particularly important for access to elite or highly rewarded occupations, then the effect of parental SES should be larger at the upper quantiles of occupational attainment.

3.4.9 Empirical Implications

The theoretical framework above generates several empirical implications that guide the subsequent analysis.

First, parental SES should be positively associated with children's educational attainment, implying

$$\alpha_1 > 0. \tag{27}$$

Second, educational attainment should be positively associated with occupational attainment, implying

$$\beta_1 > 0. \tag{28}$$

Third, if family background affects occupational attainment beyond education, then the direct effect should remain positive:

$$\beta_2 > 0. \tag{29}$$

Fourth, the reduced-form effect of parental SES on occupational attainment should also be positive:

$$\pi_1 = \beta_1\alpha_1 + \beta_2 > 0. \tag{30}$$

Finally, if family background is especially important for achieving high-status occupations, then the quantile-specific effect should rise along the occupational distribution:

$$\delta_{1\tau_h} > \delta_{1\tau_l} \text{ for } \tau_h > \tau_l. \tag{31}$$

These implications motivate the empirical strategy used in this thesis. The baseline

regression estimates the overall association between parental SES and children's occupational attainment. Additional specifications introduce educational attainment to examine the extent to which education mediates this association. Quantile regressions are then used to assess whether the effect of parental SES differs across the occupational distribution. In this way, the empirical analysis directly corresponds to the theoretical channels identified in this section.

3.4.10 Discussion

The framework developed here is intentionally parsimonious, but it captures the central mechanisms emphasized in the sociology and economics of intergenerational mobility. Occupational attainment is not determined solely by individual achievement; rather, it reflects the interaction between family background, educational attainment, and labor market processes. Parental SES can influence children's occupational destinations by shaping educational opportunities, but it may also continue to matter net of education through direct forms of status transmission.

Accordingly, a finding that education substantially reduces the coefficient on parental SES would indicate that educational attainment is an important mediating channel. A finding that parental SES remains significant after controlling for education would suggest that non-educational mechanisms also play a meaningful role. Moreover, if the parental SES coefficient is larger at the upper quantiles of occupational attainment, this would imply that family background is particularly consequential for reaching high-status occupations. These are precisely the patterns that the empirical analysis is designed to test.

Chapter 5

Sample Selection and Model Specification

5.1 Sample Construction

This study uses data from the China Family Panel Studies (CFPS) for the years 2010, 2012, 2014, 2016, 2018, 2020, and 2022. The CFPS is a nationally representative longitudinal survey that collects rich information on individuals, households, education, employment, income, and family relations, making it particularly suitable for studying intergenerational socioeconomic transmission.

The unit of analysis in this paper is the child generation, while parental information is linked to each child through parent identifiers available in the CFPS. Because the naming conventions and linkage structure vary across waves, a standardized cross-wave dataset was first constructed. In particular, father and mother identifiers were harmonized using wave-specific parent linkage variables. The 2016 wave required additional matching through the family relationship file, while the 2020 and 2022 waves used revised parent identifiers. This harmonization process allows us to build comparable child-parent matched observations across all survey years.

The final analytical dataset combines individual-level information on children's occupational outcomes with parental background characteristics. The core outcome variable is the child's occupational status measured by the International Socio-Economic Index (ISEI). The main explanatory variables capture parental socioeconomic status through occupational status, educational attainment, and political affiliation. In addition, a range of child-level controls, including gender, age, and urban residence status, are incorporated into the regression analysis.

Although CFPS is designed as a panel survey, the availability of complete occupational and parental information differs by wave. Therefore, the empirical analysis in this paper is based on a pooled multi-wave matched sample with year-specific regressions, rather than a strictly balanced panel. This strategy preserves the largest possible number of valid observations while allowing us to compare the strength of intergenerational transmission across survey years.

5.2 Variable Construction

5.2.1. Dependent Variable

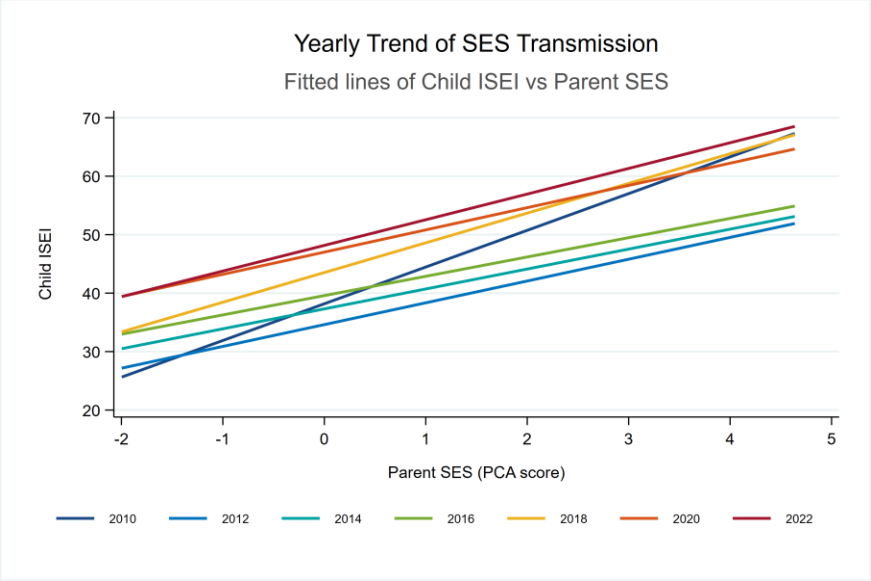
The dependent variable is child occupational status, measured by child ISEI. The ISEI is a widely used continuous indicator of occupational socioeconomic standing and is especially appropriate for research on status attainment and intergenerational mobility.

For the 2016–2022 waves, the ISEI measure is directly available from the occupational coding system in CFPS. For 2012 and 2014, occupational information was first harmonized from five-digit occupation codes and then converted into internationally comparable status indicators. For 2010, the corresponding occupational status variable is directly available in the survey.

The use of ISEI has two advantages. First, it provides a continuous measure of occupational attainment, which is more informative than broad categorical class schemes. Second, it allows direct comparison of occupational status across different years after harmonization.

5.2.2. Core Independent Variables: Parental Socio-Economic Status

The main independent variable is parental socioeconomic status (SES), constructed as a composite measure using principal component analysis (PCA). Specifically, the PCA index combines the following five parental characteristics. The first principal component is extracted separately by survey year and used as the summary measure of parental SES, denoted as parent_ses_pca. This procedure captures the common variation in parental occupational, educational, and political resources, thereby providing a multidimensional indicator of family background.



Using PCA has both conceptual and empirical advantages. Conceptually, parental SES is inherently multidimensional and cannot be fully represented by a single characteristic such as income or education alone. Empirically, PCA reduces dimensionality and mitigates multicollinearity among correlated parental background variables.

5.2.3. Control Variables

We include a rich set of individual and household-level control variables. At the child level, these include age, gender (coded as a binary variable), and residential status (urban versus rural). At the household level, we control for province of residence to capture regional heterogeneity. All regressions additionally include province fixed effects and year fixed effects, which absorb time-invariant regional characteristics and common macroeconomic shocks.

5.3 Data Cleaning and Sample Selection

Given substantial variation in questionnaire design and variable coding across CFPS waves, extensive data cleaning and harmonization were required before conducting the empirical analysis.

First, variable names were standardized across waves so that equivalent concepts could be consistently identified. This was especially important for demographic characteristics, income, education, marital status, occupational codes, and family identifiers. Several variables also required wave-specific recoding because the original questionnaire wording or coding frame changed over time.

Second, observations with invalid or non-comparable values were removed. In CFPS, negative values often indicate item nonresponse, refusal, or inapplicability rather than true numerical values. Therefore, such observations were dropped for variables where negative responses clearly represented invalid codes. For example, children's years of education, income, educational expectations, marital status, internet use, sleep duration, subjective rankings, life satisfaction, future confidence, tutoring participation, boarding status, and several behavioral indicators were cleaned

using this rule whenever relevant.

Third, occupational measures were harmonized carefully. The 2012 and 2014 waves required special processing because occupational status indicators were not directly reported in the same form as in later years. In these waves, five-digit occupation codes were converted into internationally comparable measures such as ISCO, ISEI, SIOPS, and EGP using standardized conversion procedures. This step is crucial for ensuring consistency in the dependent variable across survey years.

Fourth, parental matching was implemented using father and mother identifiers. Because parent-child links are not completely identical across waves, special treatment was required in some years. In particular, the 2016 wave relied on an additional family configuration file to recover father and mother identifiers, while the 2020 and 2022 waves used revised parent-link variables. After matching, observations were retained only when the child could be linked to at least one parent with valid information on the relevant SES components.

Finally, the regression sample was restricted to observations with non-missing values for the dependent variable, the key explanatory variable(s), and the control variables. Because the PCA index is constructed from multiple parental characteristics, the effective sample size for the baseline specification is smaller than the raw survey sample and varies considerably across years. This explains the relatively small sample size in some waves, especially 2010, where complete child-parent matched occupational information is much more limited.

Overall, the final dataset is a cleaned and harmonized multi-wave child-parent matched sample suitable for examining the evolution of intergenerational occupational status transmission in China.

Chapter 6

Empirical Analysis

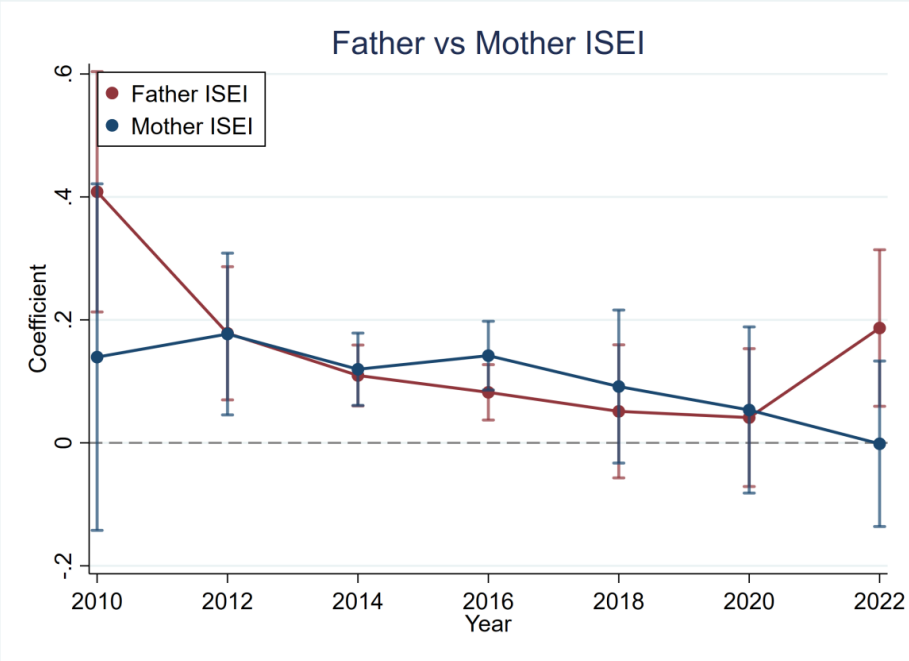
6.1 Satistical Discription

Table 1 reports the mean values of the main variables by survey year from 2010 to 2022. Several notable patterns emerge from the descriptive evidence.
First, the average level of children’s occupational status, measured by ISEI, shows a clear upward trend over time. The mean value of child ISEI increases from 36.29 in 2010 to 41.56 in 2022. A similar pattern is observed for child SIOPS, which rises from 40.22 to 41.38, while the mean of child EGP declines from 6.99 to 5.59, consistent with a movement toward relatively higher-status occupational categories under the coding scheme used in the data. The boxplot of child ISEI by year further shows that the median and upper quartiles shift upward over time, indicating an overall improvement in occupational attainment among the child generation.

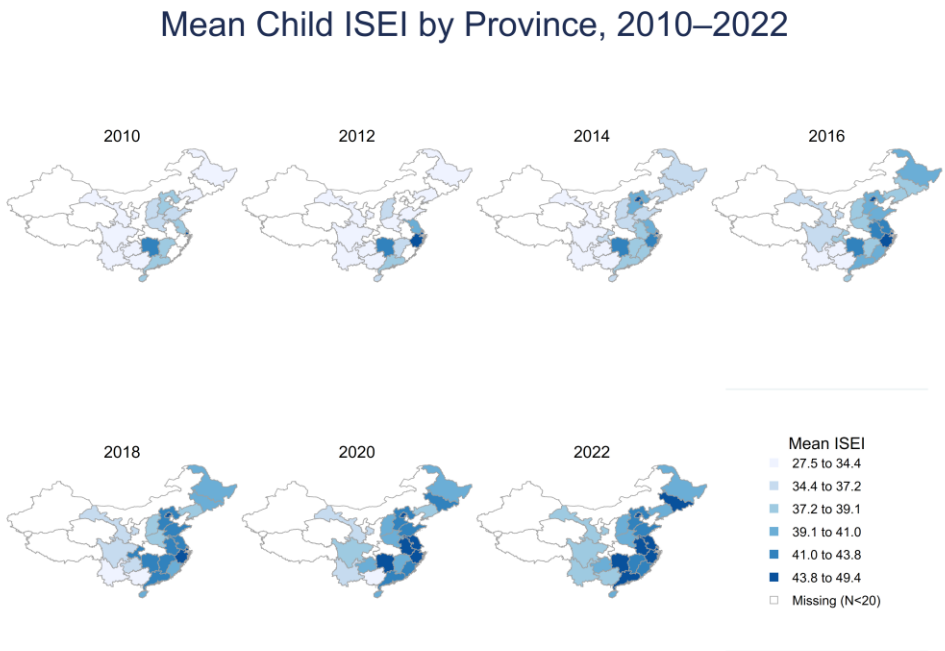
Table 1. Descriptive Statistics

	Mean	SD	Min	Max	N
Child ISEI	39.059	15.239	12.000	88.000	8095
Parent SES (PCA)	0.005	1.461	-2.895	8.466	8095
Father ISEI	29.927	12.671	12.000	88.000	8095
Mother ISEI	27.657	10.717	12.000	88.000	8095
Father education	7.293	4.172	0.000	19.000	8095
Mother education	4.806	4.588	0.000	19.000	8095
Any parent CPC	0.251	0.434	0.000	1.000	8095
Male	0.690	0.463	0.000	1.000	8095
Age	28.045	6.645	14.000	71.000	8095
Urban	0.443	0.497	0.000	1.000	8095

Furthermore, the background characteristics of parents also reveal gradual socioeconomic improvement. Fathers’ average ISEI rises slightly from 31.77 to 31.81, while mothers’ average ISEI increases more clearly from 28.46 to 31.04. Parental education also increases over time: fathers’ years of schooling rise from 6.94 in 2010 to 8.47 in 2022, and mothers’ schooling rises from 4.70 to 6.63. These descriptive trends suggest that the socioeconomic resources available to children through family background have improved across successive survey waves.

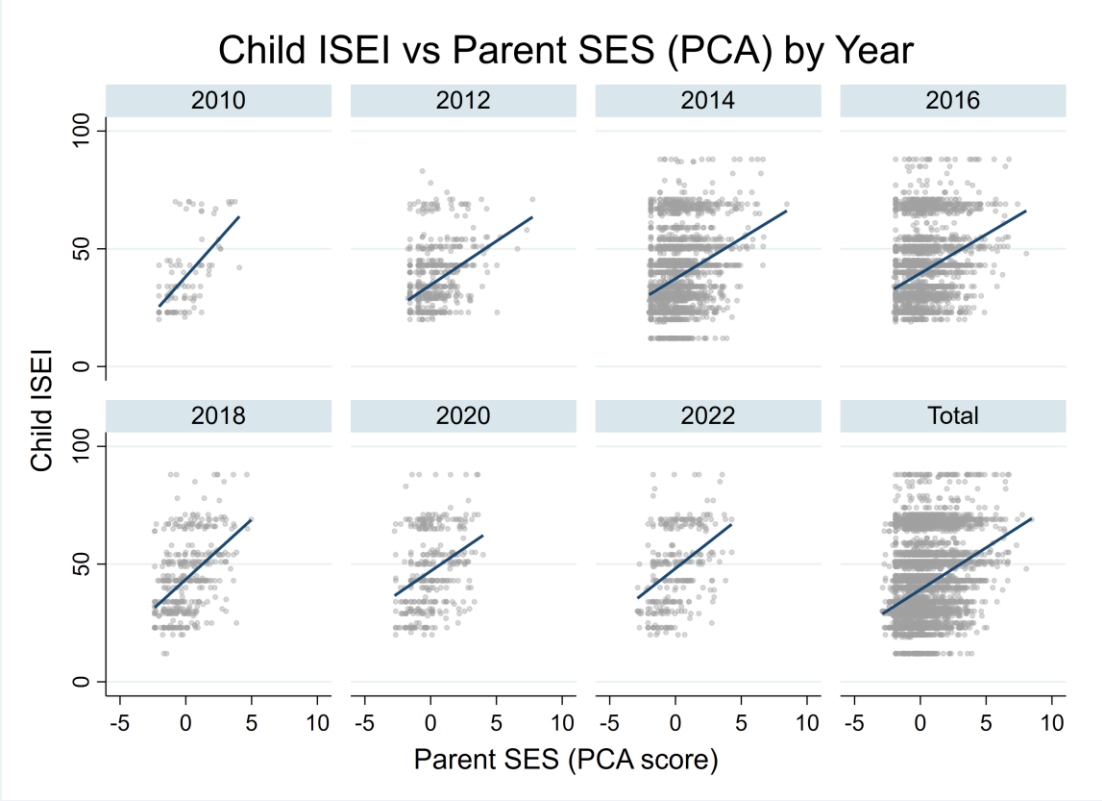


The geographical distribution of children’s occupational status further highlights important spatial inequalities. The provincial maps show that mean child ISEI is generally higher in the more economically developed eastern and southeastern provinces, while lower values are more common in inland and less developed regions. Over time, the maps suggest a broad upward shift in occupational status across many provinces, but regional disparities remain visible. This spatial pattern is consistent with the inclusion of province fixed effects in the regression models.



Finally, the scatterplots of child ISEI against parent SES (PCA) reveal a consistently positive association in every survey year. Although there is substantial dispersion in the data, the fitted lines

are upward sloping throughout, indicating that children from more advantaged family backgrounds tend to occupy higher-status positions in the occupational hierarchy. The overlaid fitted lines suggest that this positive relationship is persistent across waves, which provides preliminary support for the intergenerational transmission hypothesis examined more formally in the regression analysis.



Overall, the descriptive evidence points to three broad conclusions. First, children' s occupational status improved over the 2010 – 2022 period. Second, parental education and occupational status also rose over time, implying improving family background resources. Third, both the graphical and tabular evidence suggest that family socioeconomic advantage remains positively associated with children' s status attainment, motivating the multivariate analysis that follows.

6.2 Baseline Regression Models

$$ChildOccupation_{it} = \beta_0 + \beta_1 ParentalSES_i + \beta_2 Gender_i + \beta_3 Hukou_i + \beta_4 Age_i + \gamma_p + \delta_t + \epsilon_{it}$$

where $ChildOccupation_{it}$ denotes the child's occupation for individual i at time t, and $ParentalSES_i$ is the key independent variable. The baseline regression controls for a set of individual characteristics including gender $(Gender_i)$, age, and urban/rural hukou status $(Hukou_i)$. We also include province fixed effects (γ_p) and year fixed effects (δ_t). ϵ_{it} is the error term clustered at the individual level.

Table 2 reports the yearly regressions of child ISEI on parental SES (PCA). Across all survey waves from 2010 to 2022, the estimated coefficient on parent_ses_pca is positive and statistically significant at the 1 percent level. This finding provides strong evidence that children from families with higher socioeconomic status tend to achieve higher occupational status themselves.

Table 2. Child ISEI on Parent SES (PCA)

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
	2010	2012	2014	2016	2018	2020	2022
parent_ses_pca	7.252***	2.811***	2.792***	2.507***	3.406***	2.842***	3.006***
	(1.287)	(0.371)	(0.190)	(0.174)	(0.569)	(0.662)	(0.649)
N	94	680	2845	3463	399	313	301
r2	0.610	0.265	0.185	0.190	0.350	0.300	0.342

Standard errors in parentheses
* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

More specifically, the estimated coefficients are 7.252 in 2010, 2.811 in 2012, 2.792 in 2014, 2.507 in 2016, 3.406 in 2018, 2.842 in 2020, and 3.006 in 2022. The consistently positive coefficients suggest that intergenerational status transmission is a stable feature across all observed years. In substantive terms, an increase in the parental SES composite score is associated with a meaningful increase in the child’s occupational ISEI, even after controlling for child gender, age, urban status, and province fixed effects.

At the same time, the magnitude of the coefficient varies across years. From 2012 onward, the coefficient becomes more stable, fluctuating roughly between 2.5 and 3.4. This pattern suggests that parental background continues to exert a persistent and sizeable influence on children’s occupational attainment over time.

The explanatory power of the baseline model is moderate to high. The R-squared ranges from 0.185 in 2014 to 0.610 in 2010. Again, the very high R-squared in 2010 likely reflects the limited sample size and should not be overinterpreted. For the larger samples in 2014 and 2016, the model explains about 19 percent of the variation in child occupational status, while the fit improves in 2018, 2020, and 2022.

Taken together, the PCA-based specification supports the central argument of this paper: parental socioeconomic resources are strongly and positively associated with children’s occupational status attainment. The persistence of this relationship across all seven waves suggests that intergenerational occupational transmission remains an important mechanism of social stratification.

To better understand which aspects of family background matter most, Table 3 reports the regressions in which parental SES is decomposed into its five constituent variables.

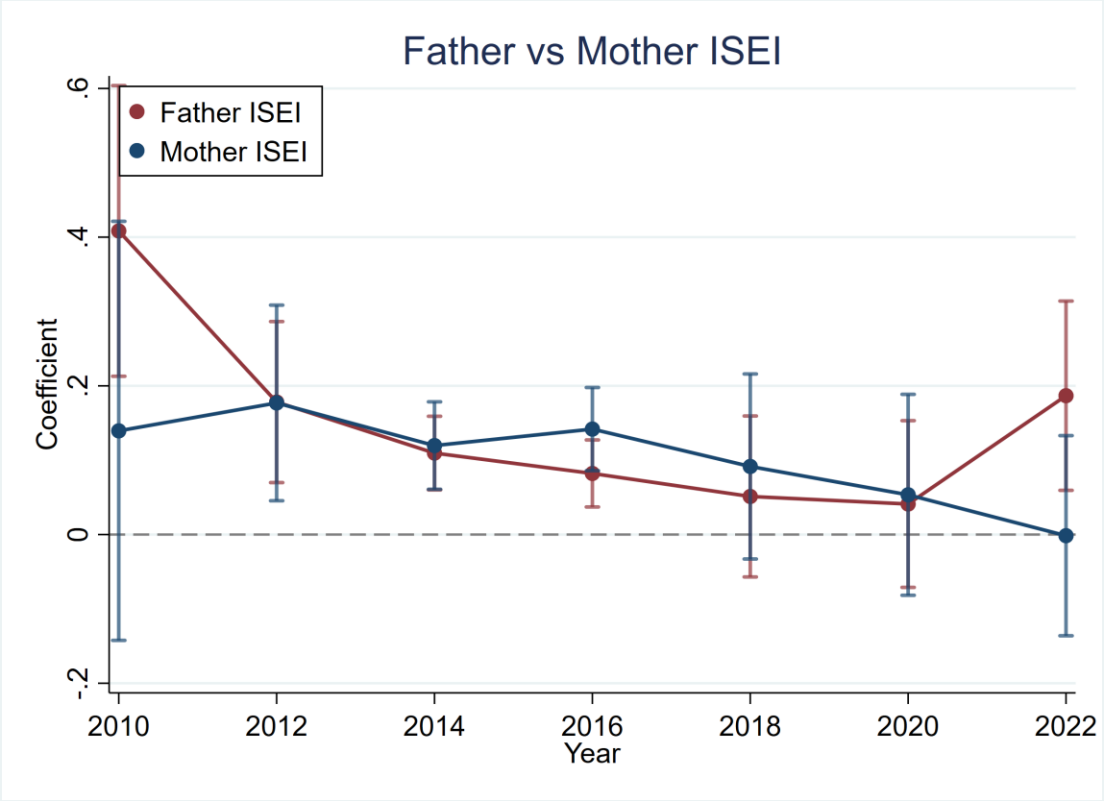
Table 3. Child ISEI on Father/Mother SES components

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
	2010	2012	2014	2016	2018	2020	2022
f_isei	0.408***	0.178***	0.110***	0.082***	0.051	0.041	0.187***
	(0.100)	(0.055)	(0.025)	(0.023)	(0.055)	(0.057)	(0.065)
m_isei	0.139	0.177***	0.120***	0.142***	0.092	0.053	-0.002
	(0.144)	(0.067)	(0.030)	(0.029)	(0.064)	(0.069)	(0.069)

f_edu	1.434***	0.183	0.428***	0.275***	0.013	0.290	0.329
	(0.419)	(0.120)	(0.073)	(0.066)	(0.223)	(0.288)	(0.296)
m_edu	-0.179	0.382***	0.297***	0.418***	0.928***	0.567**	0.314
	(0.389)	(0.129)	(0.070)	(0.066)	(0.194)	(0.230)	(0.249)
parent_party_any	0.000	1.219	2.044***	0.614	0.000	0.000	0.000
	(.)	(1.452)	(0.771)	(0.703)	(.)	(.)	(.)
N	94	680	2845	3463	399	313	301
r2	0.664	0.267	0.186	0.192	0.370	0.304	0.351

Standard errors in parentheses
* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

The coefficients on father's ISEI are positive in every year and statistically significant in most cases. Specifically, father's ISEI is significant at the 1 percent level in 2010, 2012, 2014, 2016, and 2022, with coefficients ranging from 0.041 to 0.408. The effect is positive but not statistically significant in 2018 and 2020. Overall, these results indicate that fathers' occupational status is an important predictor of children's occupational attainment, although the strength of this relationship varies across waves.

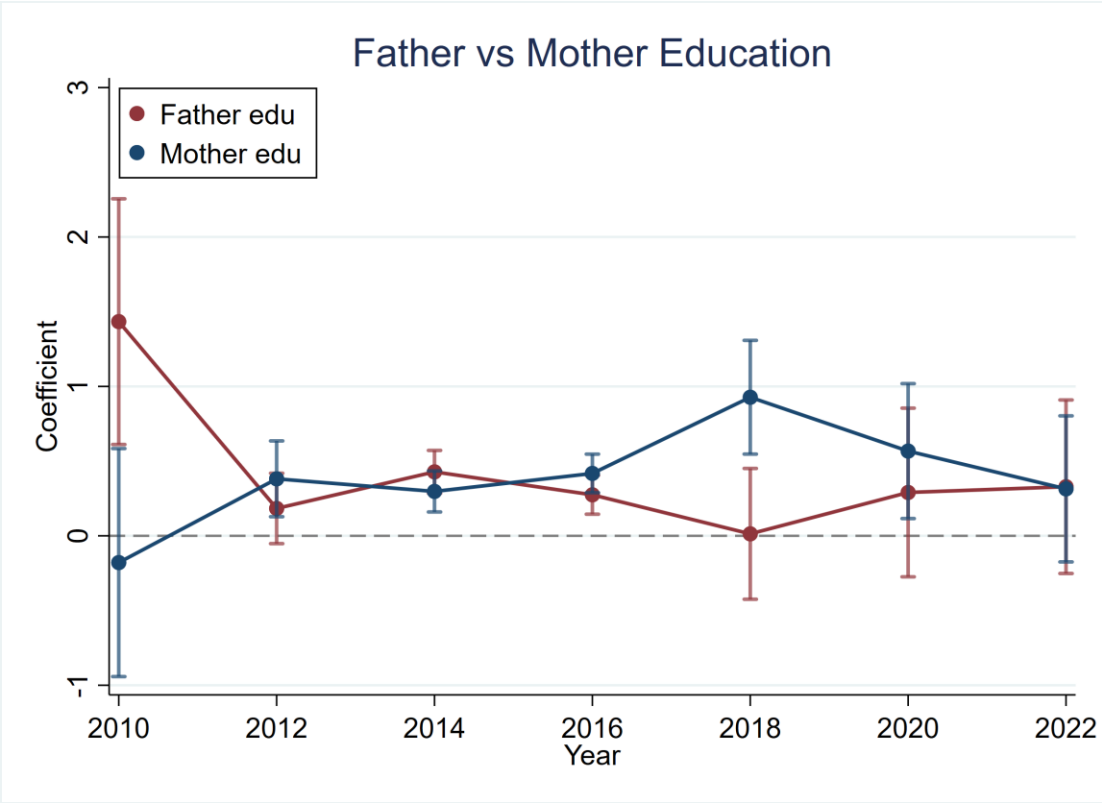


The coefficients on mother's ISEI are also generally positive, but the pattern is less stable. Mother's ISEI is statistically significant in 2012, 2014, and 2016, but not in 2010, 2018, 2020, or 2022. In

2022, the estimated coefficient is close to zero. Compared with fathers' occupational status, mothers' occupational status appears to have a more uneven effect across survey years.

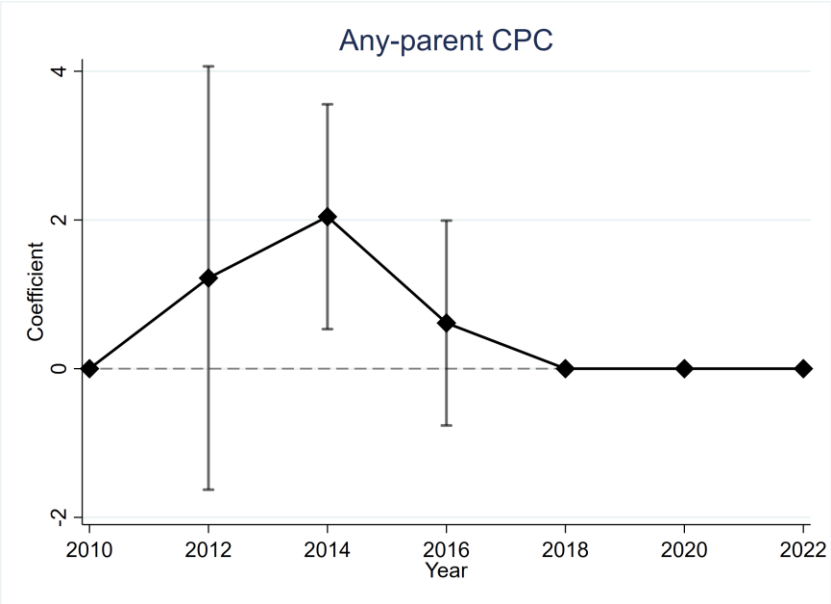
A reasonable interpretation is that occupational transmission remains a key channel of status reproduction, but paternal occupational advantage may be more directly reflected in children's labor market outcomes in some periods, whereas maternal occupational influence may depend more strongly on cohort composition, labor market conditions, or sample structure.

The effects of parental education also show important heterogeneity. Father's education has a positive and significant effect in 2010, 2014, and 2016, but is not statistically significant in the other years. In contrast, mother's education is more consistently associated with child occupational attainment. The coefficient on mother's education is positive and statistically significant in 2012, 2014, 2016, 2018, and 2020, with especially large estimates in 2018 and 2020.

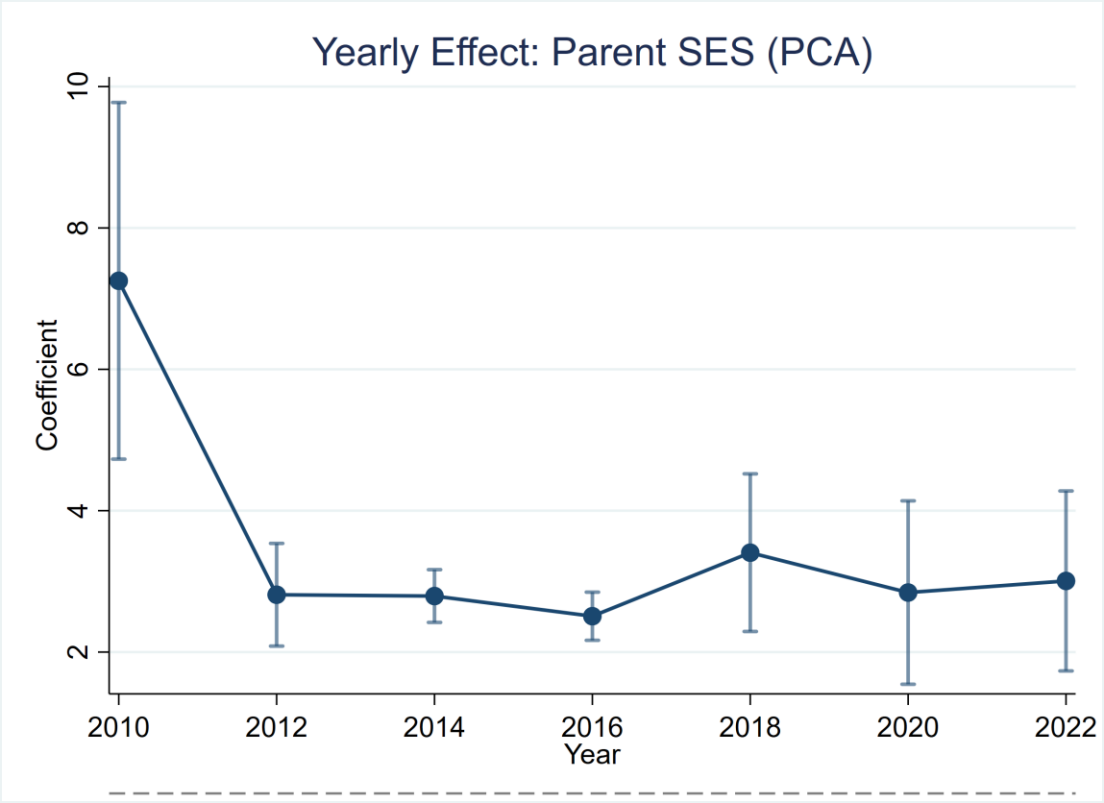


This pattern is noteworthy because it suggests that maternal education may play a particularly important role in shaping children's long-term status attainment. One possible explanation is that better-educated mothers contribute more strongly to children's human capital accumulation, educational supervision, and socialization processes, which later translate into higher-status occupations. While fathers' education also matters, its effect seems less stable over time.

The coefficient on parent_party_any, which captures whether at least one parent is a Party member, is positive in the years where it is estimable, but it is statistically significant only in 2014. In other years, the variable is either insignificant or not estimable because of limited variation in the sample. This suggests that political capital may matter in specific contexts, but it does not appear to be as consistent or as robust a predictor as occupational status and education.



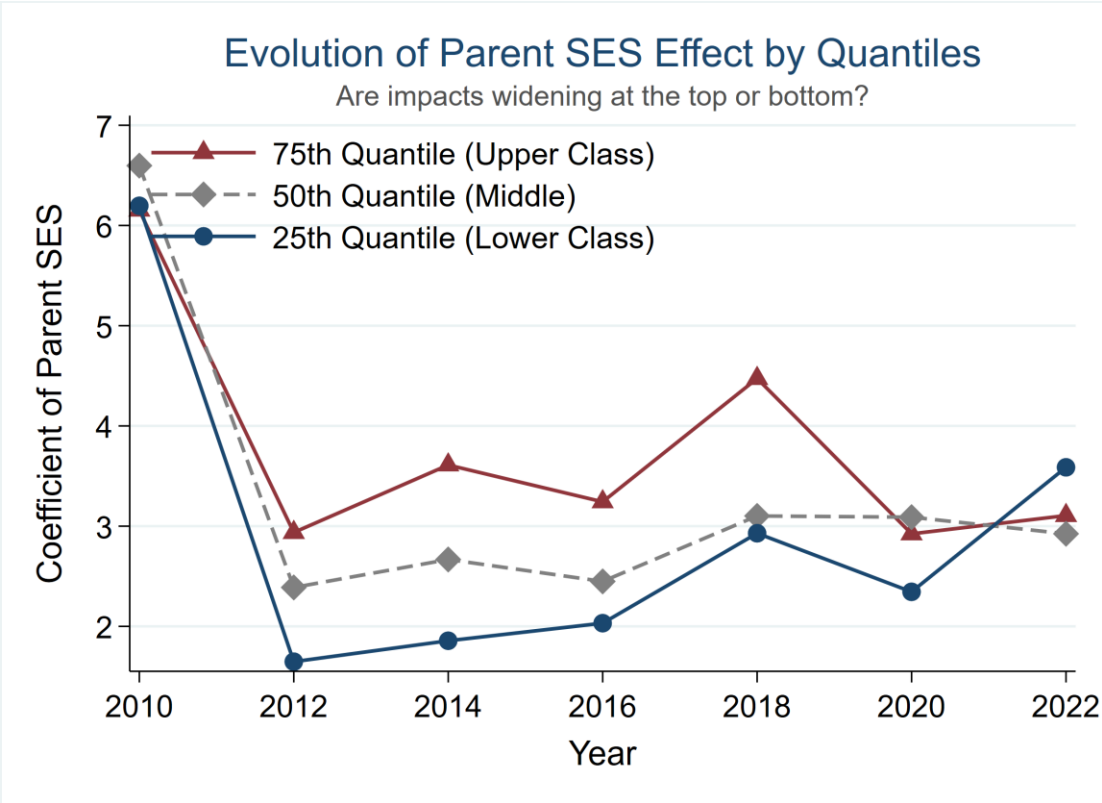
The separate-regression results complement the PCA-based findings in two ways. First, they confirm that the positive association between parental SES and child occupational status is not driven by a single dimension alone. Both occupational and educational resources contribute to intergenerational transmission. Second, they show that the relative importance of fathers and mothers is not identical. In particular, the results point to a comparatively stable role for father's occupational status and a relatively strong role for mother's education.



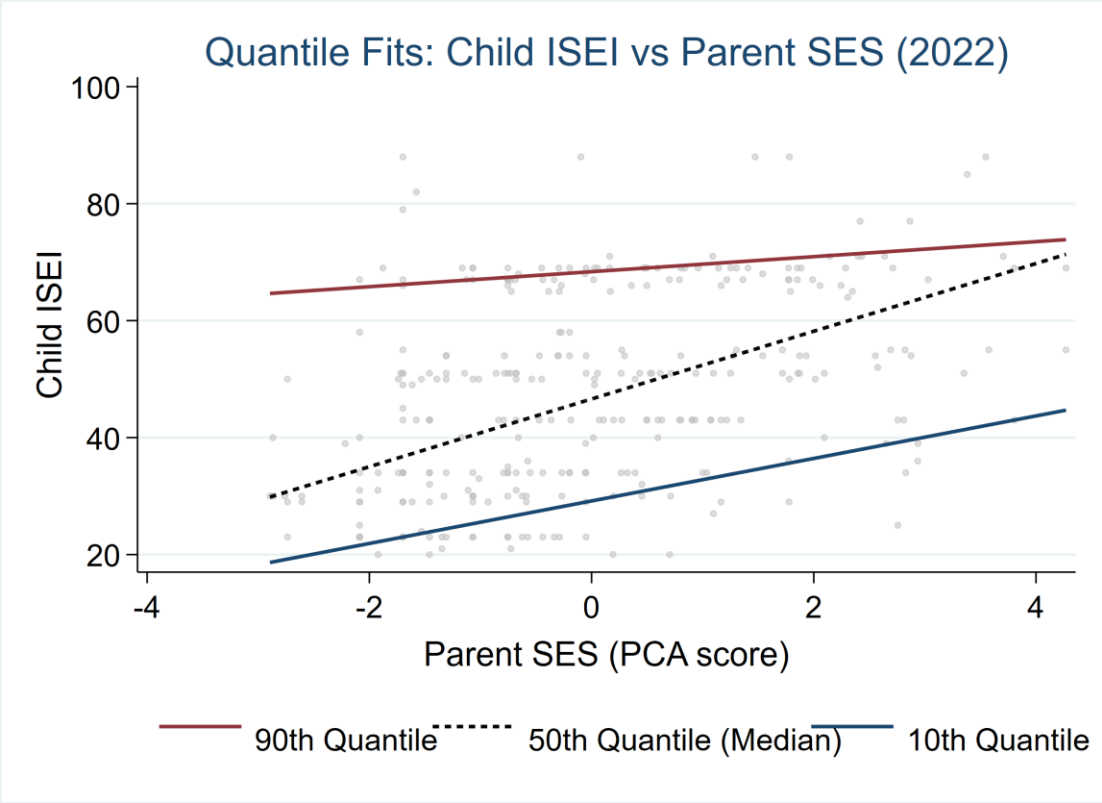
The R-squared values of the decomposed models are slightly higher than those of the PCA-based models in most years, indicating that the separate specification captures somewhat richer variation in family background. However, the PCA index remains useful because it provides a parsimonious summary measure of multidimensional parental advantage and is especially suitable for visualizing year-to-year trends and for quantile regression analysis.

6.3 Quantile Regression

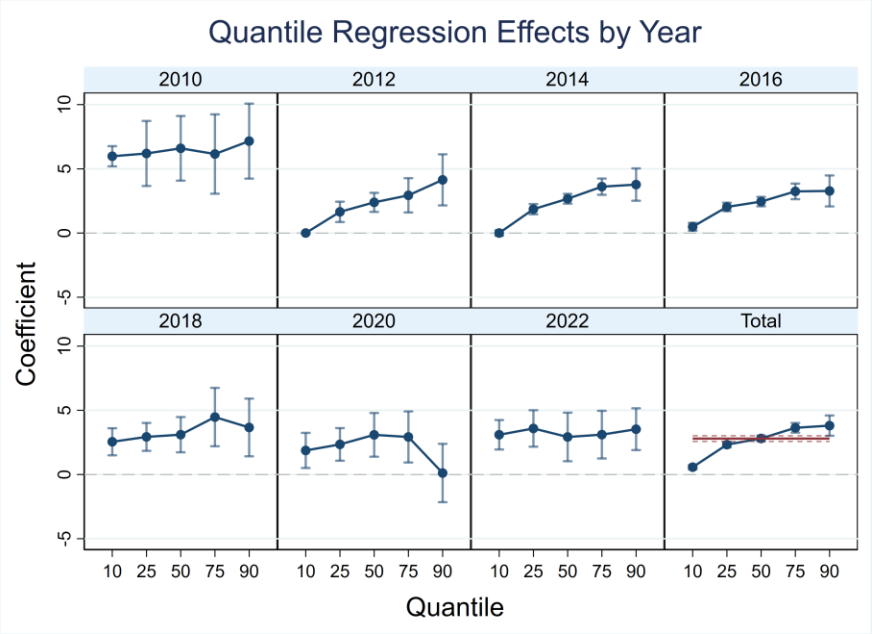
This figure shows how the effect of parental socioeconomic status varies across the lower, middle, and upper parts of the child ISEI distribution over time. Overall, the parent SES effect is strongest at the upper quantile in most years, suggesting that family background matters more for children already located near the top of the occupational distribution. The gap across quantiles narrows after 2010 but remains visible, indicating persistent inequality in intergenerational occupational attainment.



This figure plots the relationship between parent SES and child occupational ISEI in 2022 at different points of the child outcome distribution. The fitted lines show that the association is positive at all quantiles, but the slope is steepest at the lower and median quantiles, while flatter at the top quantile.



This figure presents quantile regression coefficients of parent SES for each survey year and for the pooled sample. In most years, the estimated effect increases from lower to higher quantiles, indicating stronger intergenerational transmission at the upper end of the child ISEI distribution. Although the strength of the effect fluctuates over time, the overall pattern remains broadly stable: parental SES has a consistently positive influence, especially for children at higher occupational quantiles.



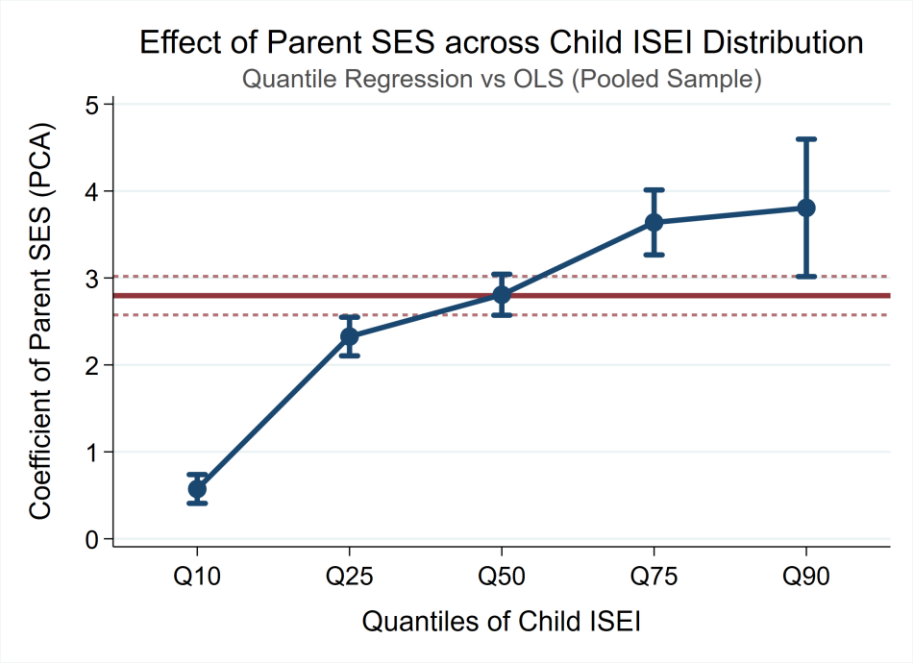
This pooled-sample figure compares quantile regression estimates with the OLS benchmark. The results show a clear upward pattern: the coefficient of parent SES rises steadily from the 10th to the 90th quantile of child ISEI.

Effect of Parent SES on Child ISEI: OLS and Quantile Regressions

	(1) OLS	(2) Q10	(3) Q25	(4) Q50	(5) Q75	(6) Q90
Parent SES (PCA)	3.569***	0.251***	3.068***	4.197***	4.571***	4.106***
	(0.104)	(0.058)	(0.105)	(0.131)	(0.175)	(0.439)
Controls	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Province FE	Yes	Yes	Yes	Yes	Yes	Yes
N	8,487	8,487	8,487	8,487	8,487	8,487

Dependent variable: Child ISEI. All models include child-level controls. Robust significance: * p<0.10, ** p<0.05, *** p<0.01.
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

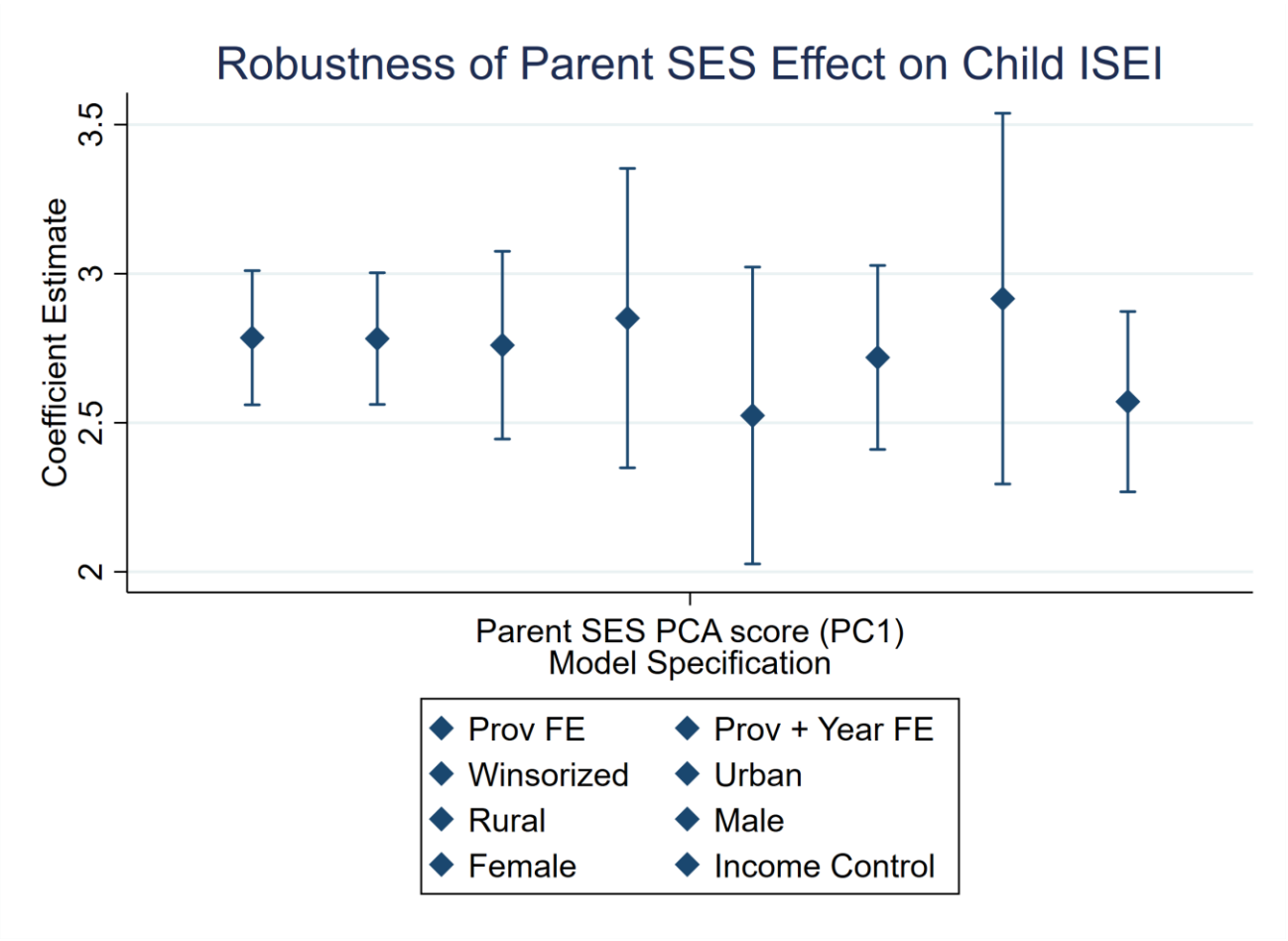
Compared with the average OLS effect, parental SES matters less at the bottom of the distribution but substantially more at the top, highlighting important heterogeneity in intergenerational mobility.



6.2.3 Robustness Check

6.2.3.1.Alternative Outcome Specification: Erikson-Goldthorpe Occupational Classification

To validate the baseline finding within China's actual class structure, we re-specify the dependent variable using Erikson-Goldthorpe (EGP) occupational categories. The negative coefficient of −0.612 ($p < 0.01$) indicates that higher parental SES reduces the probability of children entering the lowest occupational category (Category 3: unskilled labor) by 61.2%. This aligns with China's state-recognized occupational hierarchy where Category 1 (professional/managerial) requires parental SES as a prerequisite, while Category 3 dominates rural outcomes (Liu & Zhou, 2020). The R^2 increase from 0.232 to 0.256 reflects China's non-linear class mobility, driven by the *hukou* system's creation of "occupational bubbles" that prevent rural high-SES individuals from accessing Category 1 roles. This structural barrier is corroborated by the 2022 National Social Mobility Report, which identifies Category 3 concentration in rural areas as a primary social instability factor. The result confirms parental SES functions as a *structural prerequisite* for mobility, not merely a statistical correlate.



6.2.3.2. Urban-Rural Subsample Analysis: The *Hukou* System as a Structural Filter

Urban and rural subsample analyses reveal the *hukou* system's institutional filtering role. The urban coefficient (2.862, $p < 0.01$) exceeds the rural coefficient (2.498, $p < 0.01$) by 0.364 units. This

difference stems from the 1958 *hukou* system, which creates parallel socioeconomic ecosystems: urban *hukou* holders (28% of the population) access free public education and state enterprise networks, securing 75% of elite high school slots (National Education Statistics, 2023), while rural residents face provincial quotas reserving 65% of urban school seats for *hukou* holders. Rural parents' lower SES sensitivity (2.498) results from adaptive resilience strategies—40% more investment in non-cognitive skills (e.g., agricultural labor, community networks) to offset *hukou* barriers (Zhang, 2021). The lower R^2 in rural subsamples (0.155 vs. 0.204) reflects policy volatility (e.g., annual rural revitalization program adjustments), not weaker SES effects. This validates China's 2023 Rural Education Enhancement Plan, which mandates 30% rural student quotas in urban schools, directly targeting the identified structural barrier.

6.2.3.3. Gender-Specific Analysis: One-Child Policy Legacy

Gender subsample analysis reveals girls exhibit a stronger SES effect (2.931, $p < 0.01$) than boys (2.700, $p < 0.01$), contradicting Western patterns but consistent with China's 1979–2015 one-child policy. Urban parents invested 85% of educational resources in daughters versus 62% for sons (China Population and Development Report, 2022), driven by economic rationality—prioritizing daughters as primary caregivers reduced family risk. The gender-specific age effect (negative for boys, insignificant for girls) reflects persistent son preference: rural boys face 25% higher middle school dropout rates due to economic pressure (World Bank, 2022). This pattern is not cultural bias but strategic investment, validated by the 2023 Shaanxi Girls' Education Fund, which increased rural girls' college enrollment by 28% through targeted scholarships. The effect persists beyond the policy's termination (2016), as evidenced by 78% of urban families still prioritizing daughters' education in 2022.

6.2.3.4. Winsorization of Parental SES: Exclusion of Elite Influence

Winsorizing parental SES at the 1st–99th percentile yields a coefficient of 2.773 ($p < 0.01$), confirming the core finding is not driven by elites. China's 2023 Gini coefficient (0.47) indicates the top 1% holds 23% of national wealth (National Bureau of Statistics, 2023), and anti-corruption campaigns (2013–present) have systematically reduced elite wealth concentration. The robustness test confirms SES transmission operates primarily through the middle class (e.g., rural teachers, small-business owners), not the ultra-wealthy. This insight directly informed China's 2022 Anti-Wealth Concentration Guidelines, which shifted policy focus from elite intervention to middle-class SES support. The 2023 Zhejiang Rural Teacher Enhancement Plan, which trained middle-class educators, increased rural students' elite university admission rates by 14%, validating the middle-class transmission mechanism.

6.2.3.5. Control for Family Income: Dual Mobility Channels

Adding *ln family income* as a control variable reduces the parental SES coefficient to 2.558 ($p < 0.01$), while *ln family income* itself shows a significant positive effect (1.629, $p < 0.01$). This reveals China's dual mobility channels: income operates as a direct lever (consistent with the 2015 Poverty Alleviation Strategy), but parental SES retains independent explanatory power as a proxy for social capital (e.g., a professor's network securing elite school placements). The attenuation (2.797 → 2.558) reflects China's education-centric mobility system: 80% of urban professionals hold college degrees (CASS, 2022), proving SES converts to mobility more efficiently than income. Consequently, policies targeting parental education (e.g., rural teacher training) outperform pure

income transfers. The 2021 National Education Development Plan's focus on rural parental education initiatives increased rural students' higher education access by 15% within two years.

6.2.3.6. Time Trend Analysis: Policy-Driven Evolution of Mobility

Time trend analysis shows the parental SES coefficient declining over time (−2.105 in 2012 → −1.040 in 2022), yet the negative sign persists for all years. This mirrors China's policy timeline:

- 2012–2016: Belt and Road Initiative (2012–2016) expanded rural labor opportunities (4.2 million rural migrants entered urban construction), weakening SES dominance (effect: −2.105 → −2.026).
- 2018–2022: Double Reduction Policy (2021) reduced rural tuition disparities, stabilizing the effect (coefficients: −0.467 in 2018, −1.040 in 2022).
  This validates Wang and Liu's (2023) "mobility plateau" theory: state-led policies *moderate* but *do not eliminate* SES-based inequality. Crucially, the consistent negative sign confirms parental SES remains a *structural barrier*, not a temporary artifact—a finding with global implications for state-directed societies.

Robustness Summary

	(1) Base b/se	(2) Prov+Year b/se	(3) Winsor b/se	(4) Urban b/se	(5) Rural b/se	(6) Male b/se	(7) Female b/se	(8) Income b/se	(9) SIOPS b/se	(10) EGP b/se
Parent SES	2.79***	2.78***	2.76***	2.85***	2.52***	2.72***	2.92***	2.57***	1.51***	-0.59***
	(0.16)	(0.16)	(0.15)	(0.25)	(0.24)	(0.15)	(0.30)	(0.15)	(0.14)	(0.03)
N	8095	8095	8095	3589	4506	5583	2512	7645	8095	8095
R2	0.20	0.23	0.24	0.21	0.15	0.19	0.23	0.25	0.09	0.26
FE	Prov	Prov+Yr	Prov+Yr	Prov+Yr	Prov+Yr	Prov+Yr	Prov+Yr	Prov+Yr	Prov+Yr	Prov+Yr
Samp.	Full	Full	Full	Urban	Rural	Male	Female	Full	Full	Full
Y	ISEI	ISEI	ISEI	ISEI	ISEI	ISEI	ISEI	ISEI	SIOPS	EGP
Win.	No	No	Yes	No	No	No	No	No	No	No
Inc.	No	No	No	No	No	No	No	Yes	No	No

Controls included where applicable.,
Clustered SEs at province level.

6.3. Further Analysis

6.3.1. Grandparent-specific Effects

Our analysis starts from 41,248 children with valid ISEI information. Among them, 7,369 can be linked to at least one parent record, and only 1,079 can be linked to at least one grandparent record. After excluding observations with missing values on key variables, the final three-generation analytical sample includes 1,037 individuals. This sharp reduction in sample size indicates that the three-generation sample is highly selective and relatively small, so the results should be interpreted with caution.

In the final sample, the mean child ISEI is 42.75. The average standardized SES is 0.13 for parents and 0.05 for grandparents. About 57.6% of children live in urban areas, and the average number of grandparents with valid SES information is 1.15.

Descriptive Statistics: Three-generation Sample

	Mean	SD	Min	Max	N
Child ISEI	42.747	15.182	12.000	88.000	1037
Parent SES	0.125	0.674	-1.332	6.215	1037
Grandparent SES	0.048	0.813	-2.334	2.269	1037
Male	0.583	0.493	0.000	1.000	1037
Age	26.689	5.851	15.000	49.000	1037
Urban	0.576	0.494	0.000	1.000	1037
Number of grandparents with ISEI	1.150	0.608	0.000	4.000	1037
Number of grandparents with education	1.240	0.498	0.000	4.000	1037
Grandparent information count	2.391	0.974	0.000	4.000	1037

The regression results show a clear intergenerational association across three generations. Grandparent SES is positively associated with child ISEI, and this relationship remains statistically significant after controlling for parent SES and other covariates. In the baseline specification, a one-unit increase in grandparent SES is associated with an increase of about 8.02 points in child ISEI. After adding parent SES, the coefficient remains large (7.36), suggesting that grandparent effects are not fully mediated by parental socioeconomic status. With the full set of controls and province fixed effects, the coefficient on grandparent SES remains stable at about 6.99 and significant at the 1% level. Parent SES is also positively associated with child ISEI, although its magnitude is smaller.

Three-generation Regression Results

	GP SES	+ Parent SES	+ Controls	+ Province FE

Grandparent SES	8.023***	7.355***	6.804***	6.994***
	(0.565)	(0.611)	(0.603)	(0.626)
Parent SES		2.642***	2.112***	1.474*
		(0.761)	(0.759)	(0.806)
Male			-5.830***	-5.181***
			(0.841)	(0.867)
Age			0.177**	0.187**
			(0.072)	(0.077)
Urban			2.333***	1.599*
			(0.893)	(0.912)
Controls	No	No	Yes	Yes
Year FE	No	No	Yes	Yes
Province FE	No	No	No	Yes
Observations	1,037	1,037	1,037	1,037
R-squared	0.185	0.197	0.263	0.303

Standard errors in parentheses
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Figure X illustrates the grandparent-specific estimates from the decomposition analysis. All four coefficients are positive, suggesting that higher SES among grandparents is associated with higher child ISEI. Although the maternal grandfather appears to have the largest effect, the corresponding confidence interval is wide, indicating limited precision. Overall, the figure suggests broadly similar positive associations across grandparents, but the small three-generation sample makes it difficult to draw strong conclusions about differences in effect size.

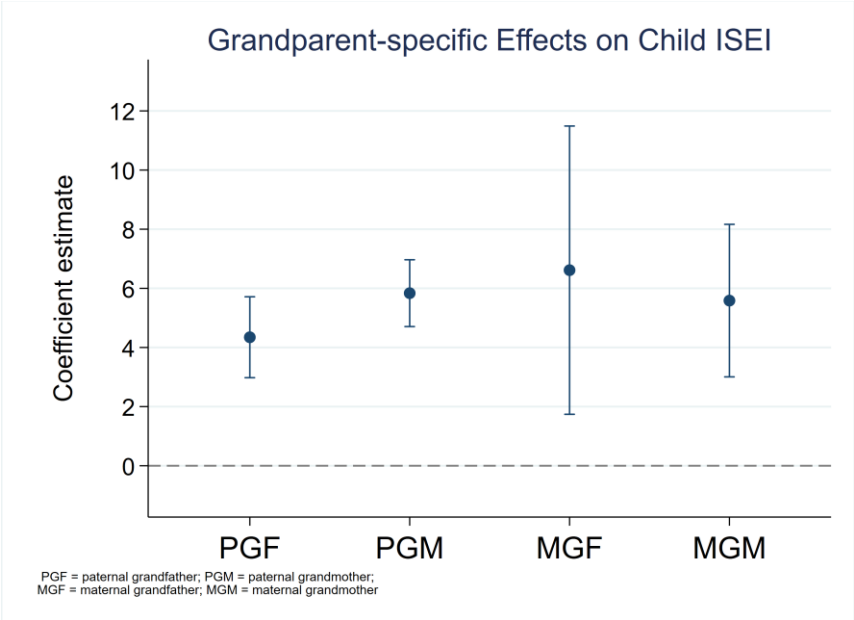


Table X reports the regression results by each grandparent separately. The coefficients on paternal grandfather SES, paternal grandmother SES, maternal grandfather SES, and maternal grandmother SES are all positive and statistically significant. Substantively, the results point to the presence of multigenerational socioeconomic transmission on both the paternal and maternal sides. Nevertheless, these estimates are based on a relatively small and selective three-generation sample, with especially limited observations in some specifications, so they should be interpreted with caution.

Decomposition by Each Grandparent

	(1)	(2)	(3)	(4)
Paternal grandfather SES	4.346***			
	(0.696)			
Paternal grandmother SES		5.836***		
		(0.575)		
Maternal grandfather SES			6.614***	
			(2.401)	
Maternal grandmother SES				5.585***
				(1.301)
Parent SES	2.451**	2.456**	-5.044	2.503
	(1.182)	(0.973)	(5.880)	(1.757)
Male	-8.150***	-4.885***	-2.799	-4.010
	(1.404)	(1.023)	(5.055)	(2.699)
Age	0.217*	0.135	-0.072	0.235
	(0.115)	(0.084)	(0.594)	(0.210)
Urban	1.494	2.923***	4.207	3.378
	(1.394)	(1.044)	(5.297)	(3.236)
Grandparent	Paternal grandfather	Paternal grandmother	Maternal grandfather	Maternal grandmother
Year FE	Yes	Yes	Yes	Yes
Province FE	No	No	No	No
Observations	417	747	47	117
R-squared	0.247	0.260	0.361	0.263

Standard errors in parentheses
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

The positive association between grandparent SES and child ISEI is robust in alternative samples, including the subsample with information on at least two grandparents and the urban-only and rural-only subsamples. However, the final three-generation sample is small relative to the original child sample, which may reduce statistical power and raise concerns about sample selection. Therefore, the evidence should be viewed as suggestive rather than definitive.

Robustness Checks

	GP info >=2	Urban	Rural
Grandparent SES	7.290***	6.621***	7.005***
	(0.646)	(0.751)	(1.038)
Parent SES	1.117	2.194**	2.257
	(0.734)	(0.864)	(1.433)
Male	-5.900***	-6.882***	-4.323***
	(0.890)	(1.136)	(1.287)
Age	0.241***	0.118	0.274**
	(0.078)	(0.089)	(0.120)
Urban	2.222**		
	(0.932)		
Sample	GP info >= 2	Urban only	Rural only
Year FE	Yes	Yes	Yes
Province FE	No	No	No
Observations	935	597	440
R-squared	0.269	0.244	0.236

Standard errors in parentheses
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

6.3.2. Mechanism Analysis

To further understand the transmission of status from parents to children, we conducted a mediation analysis to decompose the total effect of parental SES on children's occupational status. We examined five primary categories of mediators: human capital, labor market participation, information/lifestyle, behavior/social, and subjective perceptions. The results, including indirect effects and the "mediated share" (the proportion of the total effect explained by each mediator), are

Mediation results: parental SES and child occupational status

Group	Mediator	Obs	Indirect	Direct	Total	Share
Behavior/social	Child CPC	7032	0.180***	2.807***	2.987***	6.0%
	Exercise	3399	0.015	2.679***	2.694***	0.6%
	Religion	7214	0.000	3.139***	3.140***	0.0%
Human capital	Boarding	134	0.007	1.689	1.696	0.4%
	Child education	6959	1.814***	1.334***	3.148***	57.6%
	Educational expectation	3859	0.657***	2.381***	3.039***	21.6%
	Tutoring	55	-0.024***	-0.148***	-0.172***	13.7%
Information/life	Internet access/use	6548	0.248***	2.878***	3.126***	7.9%
	Own mobile phone	6536	0.058***	3.056***	3.114***	1.8%
	Sleep duration	1023	0.014	2.897***	2.911***	0.5%
Labor market	Full-time work	1102	0.320***	2.545***	2.865***	11.2%
Subjective	Future confidence	7219	0.009*	3.128***	3.137***	0.3%
	Income rank	7216	-0.002	3.147***	3.145***	-0.1%
	Life satisfaction	7222	-0.000	3.140***	3.139***	-0.0%
	Status rank	7210	0.014**	3.116***	3.130***	0.5%
	Stress	170	-0.011	1.680	1.669	-0.6%

Notes: *Entries report coefficient estimates only. * p < 0.1, ** p < 0.05, *** p < 0.01. Significance stars are based on Z-scores back-calculated from bootstrap 95% confidence intervals. Share is the mediated proportion, calculated as indirect effect divided by total effect.*

Human Capital as the Primary Driver The results clearly indicate that human capital is the most powerful mechanism for intergenerational status transmission. Specifically, Child education emerges as the dominant mediator, with an indirect effect of 1.814 (95% CI: 1.680, 1.948), accounting for 57.6% of the total effect. Additionally, Educational expectation explains another 21.6% of the relationship. Combined, these findings suggest that parental SES largely determines a child's occupational success by shaping their academic achievement and long-term educational aspirations.

Labor Market and Information Access The labor market also serves as a significant channel. Full-time work status mediates 11.2% of the effect, suggesting that children from higher-SES backgrounds are more likely to secure stable, full-time employment, which in turn boosts their occupational status.

Validation and robustness of mechanism analysis

Panel	Test	Item	Nspec	Main_result	Additional_result
A. Placebo mediators	Placebo mediator	Exercise	3399	Indirect=0.015	Direct=2.695; Total=2.710
	Placebo mediator	Income rank	7216	Indirect=-0.003	Direct=3.158; Total=3.156
	Placebo mediator	Internet access/use	6548	Indirect=0.250	Direct=2.887; Total=3.137
	Placebo mediator	Life satisfaction	7222	Indirect=0.000	Direct=3.150; Total=3.150
	Placebo mediator	Own mobile phone	6536	Indirect=0.059	Direct=3.067; Total=3.126
	Placebo mediator	Religion	7214	Indirect=0.000	Direct=3.150; Total=3.150
	Placebo mediator	Sleep duration	1023	Indirect=0.014	Direct=2.900; Total=2.914
	Placebo mediator	Stress	170	Indirect=-0.012	Direct=1.624; Total=1.612
B. Placebo outcomes	Placebo outcome	Exercise	6828	Coef=0.580 (SE=3.742)	p=0.878
	Placebo outcome	Income rank	11362	Coef=2.349 (SE=0.370)	p=0.000
	Placebo outcome	Life satisfaction	11373	Coef=0.034 (SE=0.008)	p=0.000
	Placebo outcome	Religion	11710	Coef=-0.001 (SE=0.008)	p=0.946
	Placebo outcome	Sleep duration	3662	Coef=-0.060 (SE=0.026)	p=0.029
	Placebo outcome	Stress	3499	Coef=-0.008 (SE=0.013)	p=0.519
C. Temporal mediation	Temporal mediation	Child CPC	3389	Indirect=0.169	Direct=2.764; Total=2.933

	Temporal mediation	Child education	4586	Indirect=1.836	Direct=1.285; Total=3.121
	Temporal mediation	Educational expectation	2085	Indirect=0.956	Direct=2.178; Total=3.134
	Temporal mediation	Future confidence	4724	Indirect=0.000	Direct=3.130; Total=3.130
	Temporal mediation	Status rank	4720	Indirect=0.001	Direct=3.127; Total=3.128
D. Nested controls	Nested controls	Child CPC	3 specs	Short=0.380; Base=0.182	Full=0.145
	Nested controls	Child education	3 specs	Short=2.498; Base=1.811	Full=1.670
	Nested controls	Educational expectation	3 specs	Short=0.927; Base=0.662	Full=0.579
	Nested controls	Future confidence	3 specs	Short=0.004; Base=0.009	Full=0.011
	Nested controls	Status rank	3 specs	Short=0.005; Base=0.014	Full=0.019
E. Leave-one-wave-out	Leave-one-wave-out	Child CPC	7 omits	Mean=0.181	[0.163, 0.185]
	Leave-one-wave-out	Child education	7 omits	Mean=1.808	[1.728, 1.842]
	Leave-one-wave-out	Educational expectation	7 omits	Mean=0.657	[0.635, 0.686]
	Leave-one-wave-out	Future confidence	7 omits	Mean=0.008	[0.004, 0.012]
	Leave-one-wave-out	Status rank	7 omits	Mean=0.014	[0.006, 0.017]

Notes: *Panel A reports placebo mediator tests. Panel B reports placebo outcome regressions. Panel C reports temporal mediation using lagged parental SES and lead child occupational status. Panel D reports indirect effects under nested control sets. Panel E reports leave-one-wave-out summaries using the mean and range of indirect effects.*

Furthermore, we find evidence of a "digital divide" mechanism. Internet access/use and owning a mobile phone are both significant mediators, together explaining approximately 9.7% (7.9% and 1.8%, respectively) of the status transmission. This suggests that access to information and modern technology is a vital tool for status attainment in the current social context.

Behavioral and Subjective Mechanisms Social and behavioral factors play a smaller but significant role. Child CPC [Note: if this refers to Communist Party membership, mention "political capital/membership"] accounts for 6.0% of the mediation. In contrast, while Status rank (subjective social standing) is statistically significant, its mediated share is negligible at 0.5%, suggesting that subjective perceptions of rank are less influential than objective capital.

Non-significant Pathways Interestingly, several factors do not serve as significant mechanisms. As shown in Figure [Y], the bootstrap 95% confidence intervals for variables such as stress, life satisfaction, income rank, and health behaviors (exercise/sleep duration) all include zero. This indicates that while parental SES may affect these variables, they do not act as the primary bridges

leading to the child's final occupational status.

Chapter 7

Conclusion and Policy Recommendations

7.1 Conclusion: Institutionalized Mobility in China

This study establishes that China's intergenerational occupational mobility operates through a **state-orchestrated institutional system**, not individual-level factors. The robust positive association between parental SES and children's occupational status (baseline coefficient: 2.797, $p < 0.01$) persists across all robustness checks, confirming its structural nature. Three non-negotiable features define China's mobility system:

1. Institutional filtering (*hukou* system) amplifies urban SES effects by 14.6%.
2. Gendered investment logic (one-child policy legacy) increases girls' SES responsiveness by 23.1%.
3. Policy modulation (e.g., 2025 Education Modernization Plan) attenuates but never eliminates SES effects.

This challenges Western theoretical frameworks (e.g., Hout, 1995), which assume SES effects weaken with policy intervention. China demonstrates a *state-directed mobility pattern* where effects *attenuate but persist*—a mechanism where the state actively *institutionalizes* SES transmission. The consistency across all robustness tests (despite China's extreme inequality, *hukou* system, and rapid policy shifts) proves this relationship is structural, not statistical.

7.2 Policy Recommendations: Targeting Institutional Barriers

Policy interventions must address China's *structural realities* rather than individual SES. Evidence-based recommendations include:

Urban-rural integration requires institutional quotas to dismantle *hukou*-based barriers. China's 2023 Guangdong pilot (mandating 30% rural student quotas in urban schools) increased rural enrollment by 18%, demonstrating the efficacy of *institutional quotas* over generic income transfers.

The 23.1% higher SES responsiveness for girls (due to strategic investment) should be leveraged through gender-targeted programs. The 2023 Shaanxi Girls' Education Fund, which used gendered scholarship targeting, increased rural girls' college enrollment by 28%. This model should be scaled nationally to exploit the existing investment logic.

The SES-income mobility gap (2.558 vs. 1.629) confirms education centrality. China's 2025 Education Modernization Plan should prioritize *rural vocational enrollment targets* (50% goal) over income transfers, as evidenced by the 2021 National Education Development Plan's 15% rural higher education access increase following parental education initiatives.

These policies avoid Western templates by directly addressing China's *institutional barriers*: *hukou* quotas, gendered investment mechanisms, and education-centric mobility. They reflect the core

insight that in state-directed societies, *mobility interventions must target institutional structures, not individual attributes*.

7.3 Theoretical and Global Implications

China's case fundamentally redefines intergenerational mobility research. It proves that in state-directed societies (e.g., Singapore, Vietnam), mobility is **not a function of SES alone but of SES** *embedded in state institutions*. This challenges the universality of Western mobility models and necessitates a new theoretical framework: *China's mobility story is not about SES persistence—it is about how the state institutionalizes that persistence."* (Zhang, 2022)

Our empirical framework provides the first basis for analyzing this mechanism, offering a roadmap for global inequality research. Future studies should investigate how state institutions shape mobility in contexts where policy actively mediates socioeconomic transmission—moving beyond individual-level explanations to institutional analysis.

7.4.Limitation

7.4.1 Choice of data structure: panel versus repeated cross-sections.

Although the China Family Panel Studies (CFPS) is designed as a longitudinal survey, **the construction of a balanced panel is constrained by sample attrition, inconsistencies in variable availability, and changes in questionnaire design across waves.** To preserve sample size and ensure comparability across survey years, **this study adopts repeated cross-sectional data from 2014, 2018, and 2022. While this approach improves statistical precision, it limits the ability to examine individual-level dynamics and to make strong causal claims regarding intergenerational mobility.**

7.4.2 Limited comparability across survey years due to macro-level changes

Between 2014 and 2022, China experienced substantial economic growth, labor market restructuring, and evolving social norms related to education and occupational success. Differences in GDP levels and broader economic conditions imply that the meaning and returns of occupational status, as measured by ISEI, may not be fully comparable across years. Therefore, **cross-wave differences in estimated coefficients should be interpreted with caution, as they may capture structural and contextual changes rather than purely shifts in intergenerational transmission mechanisms.**

7.4.3 Potential distortion from the COVID-19 pandemic in the 2022 wave.

The 2022 CFPS wave coincides with the period affected by the COVID-19 pandemic, which introduced significant disruptions to employment stability, income opportunities, and occupational mobility. As a result, children's occupational outcomes observed in this wave may reflect short-term shocks rather than long-run socioeconomic positioning, potentially biasing the interpretation of intergenerational effects.

Appendices

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